A Critical Assessment of Libertarian Paternalism

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Abstract

1 Introduction and Motivation

Libertarian paternalists claim to have managed to reconcile libertarianism and paternalism. In this article I discuss whether this claim is valid.[1]

As libertarians and paternalists look at the individual as decision-maker from radically different perspectives, libertarian paternalists are self-consciously aware both of the logical feat they are trying to pull off (one of their flagship articles is called '*Libertarian Paternalism Is not an Oxymoron*'), and of the importance of what they are attempting to achieve.[2]

What makes their approach so interesting? Traditional paternalists have always emphasized the inability of the individual to choose what is *instrumentally* good either for herself or for society (the 'inefficient-means' paternalistic justification), or *intrinsically* good in the light of some idea of what the proper goals a society or an individual should pursue (the 'undesirable-ends' justification).[3]

In the twentieth century, however, paternalists have found themselves on the defensive. As far as the undesirable-ends justification is concerned, contemporary western society is probably more ill at ease than it has ever been

[1] For the sake of brevity, in this paper I often refer to Thaler and Sunstein (2008) as *Nudge*, and to Rebonato (2012) as *Taking Liberties*.

[2] 'The twentieth century is pervaded by a great deal of artificial talk about the possibility of a "Third Way". We are hopeful that libertarian paternalism offers a real Third Way – one that can break through some of the least tractable debates in contemporary democracies', *Nudge*, Chapter 18, page 252.

[3] The distinction between means-paternalism and ends-paternalism is clearly discussed in New and Le Grand (forthcoming). This important distinction is not commonly made in the literature about paternalism, but is of great importance to my discussion of libertarian paternalism. Libertarian paternalists would probably position themselves at the 'means' end of the spectrum. I intend to show in what follows that what they propose is far more tilted towards 'ends-paternalism' than it is often recognized.

with the idea of a universally-acknowledged superior good that a mandarin class should impose on the citizenship. And the inefficient-means justification of paternalism has come under attack from the impressive analytical framework that goes under the generic rubric of decision theory (often as adopted by the neoclassical-economics consensus). This approach finds its culmination in the concept of Pareto efficiency, and in the two Welfare theorems.[4][5]

If a programme of paternalistic inefficient-means-based intervention is still advocated, there are therefore three main logical routes open to twenty-first-century paternalists: either they must claim that real-life conditions are too far removed from the idealized requirement of the theorems for the results to apply; or they must limit the scope of paternalistic intervention to those rather special situations (such as externalities) where unfettered choices of rational, well-informed individuals do not bring about optimal results; or they have to 'go for broke' and try to strike a fatal blow to the very foundations of rational decision theory.

The latter is indeed the route implicitly taken by the libertarian paternalists: they refuse, in fact, even to start the intellectual journey built on the rational features of *Homo Economicus.* They do so by denying the validity – either as useful approximations or in an *as-if* sense – of the most basic axioms about how individuals make choices. Indeed, given the importance accorded by libertarian paternalists to *status quo* bias and framing in general, even transitivity of preferences is not guaranteed,[6] and the whole edifice that builds up to expected utility maximization falls to pieces. The important point to stress here is, according to libertarian paternalists, what individuals freely choose is not only not always good for society, *but even for them.*

So much for what the libertarian paternalists do *not* accept. But how do libertarian paternalists go about building the positive part of their programme? The answer in not easy, because libertarian paternalism is a proteiform beast, and many of the initiatives that have been presented seem at times to be linked by a tenuous link. In order to avoid confusion, let me therefore offer a narrow, but reasonably precise, definition of libertarian paternalism as follows.

> Libertarian paternalism is the set of interventions aimed at overcoming the unavoidable cognitive biases and decisional inadequacies

[4] For two clear derivations at undergraduate and graduate level, repsectively, see Campbell et al (2008) and Kreps (1990). For a discursive critical discussion that focusses on the conceptual

implications of Pareto efficiency, of the two welfare theorems and of the conditions under which they obtain, see, eg, Marglin, (2008), Appendix A.

[5] When the existence of market failures such as externalities is recognized, economic theory suggests that the 'neat' solution is simply to introduce the 'missing market', and then let individuals free again to choose without hindrance.

[6] See Gilboa (2010), for a clear and precise, yet non-technical, discussion of the axioms of choice, or Appendix B of the same work for a simple mathematical treatment. (Gilboa (2010), available at http://mitpress.edu/rationalchoice). For an example of the libertarian paternalistic assertion that transitivty of choice is often violated, see, eg, Thaler and Sunstein (2003): "People do not exhibit rational expectations, fail to make forecasts that are consistent with Bayes' rule, [...], *exhibit preference reversals* (that is they prefer A to B and B to A) and make different choices depending on the wording of the problem...' [emphasis added].

> of an individual by exploiting them in such a way as to influence her decisions (in an easily reversible manner) towards choices that she herself would make if she had at her disposal unlimited time and information, and the analytic abilities of a rational decision-maker (more precisely, of *Homo Economicus*).[7]

Of course, the fair thing to do here would be to present the libertarian paternalists' own definition of what their programme is about, rather than mine. The problem is that it is difficult to find in their literature a single concise definition that addresses all the important aspects (the "necessary and sufficient conditions") of libertarian paternalism. I have therefore taken the liberty to provide a distillation of what I understand the necessary and sufficient conditions to be for a proposal to be libertarian paternalistic. This is a subset of what the literature and commentators refers to as libertarian paternalism, but, in my view, the *only* truly original part of the programme – the only part, that is, that can stake a reasonable claim to being 'the new Third Way'. It is the libertarian paternalism I deal with in this paper.

1.1 How Can One Possibly Worry?

In contemporary political and philosophical debate, it matters a lot whether a programme of interventions is paternalistic or not. If it is, the proponent of the paternalistic measure usually feels obliged to provide lengthy justifications as to why it should be admissible to interfere with an individual's autonomy of choice.

Libertarian paternalists seem to dismiss with an impatient wave of the hand most of these exculpatory concerns. Indeed, they adopt a philosophically 'muscular' and no-nonsense attitude when they state that '[t]he first misconception is that there are viable alternatives to paternalism'[8] and they urge us to 'abandon the less interesting question of whether to be paternalistic or not, and turn to the more constructive question of how to choose among the possible choice-influencing options'[9]. The reason for the their failure seriously to engage with the question of how paternalism should be justified can be found in the special version of their paternalism, whose interventions must be *easily reversible*. If they truly are, the argument goes, why should one be overly concerned with the question of whether we label them as paternalistic? And if we accept that easy reversibility 'justifies' paternalism without the need for further discussion, shouldn't the examination of the acceptability of a libertarian paternalistic intervention be very easy indeed, and limited to checking that the choices are indeed easily reversible?

[7] A note on the use of personal pronouns: I have arbitrarily chosen to use the masculine pronouns when referring to choice architects and libertarian paternalists and the feminine pronouns when referring to the individuals making the choices. Nothing deep should be read into this convention. However, when referring to *Homo Economicus*, I have always used the masculin pronoun.

[8] Thaler and Sunstein, (2003), page 175

[9] *Nudge*

I intend to explain in this paper why I think that the 'case for the defence' based on the reversibility argument is not as open-and-shut as it seems, but an example can already highlight some of my concerns. Almost 100% of Austrians have agreed to being organ donors, but only 12 % of Germans have.[10] Trying to account for this enormous difference by looking, say, at cultural, social or medical-institutional differences between the two countries would be an almost complete waste of time. The simple explanation is that 'Germans must opt in, whereas the Austrians are asked whether they do *not* want to become organ donors'. [11] Here we have an example of a cheap, effective, and, in principle, *easily reversible*, nudge: the choice of a default option. Nobody can deny that, if an Austrian citizen objected to being a potential organ donor, the cost associated with reversing the default would be very low. Conversely, I can imagine that it must be easy for a German to put her name on an organ-donor register. Both Germans and Austrians can therefore exercise their *nominal* freedom of choice at very little cost. In practice, however, the figures above suggest that they do not. And they do not exactly for the cognitive reasons (say, *status quo* bias, decisional inertia) that the libertarians paternalists are so good at identifying and at exploiting for their programme. What has become, then, of their 'freedom of choice'? This example suggests that looking at *effective*, rather than *nominal*, freedom of choice might be a better indication of whether a measure can be called 'libertarian' or not.

Let us put the issue of reversibility to one side. Perhaps we may reach the conclusion that the libertarian paternalists are not quite as libertarian as they would like to be. Still, their 'softer' version of paternalism would at first blush appear more palatable (eg, less intrusive, less coercive) than the 'hard' version. But is it? I will argue that, exactly because so 'discreet', their version of paternalism can, paradoxically, be less desirable than the 'hard' variety'. I can adduce at least two reasons for this statement.

First, exactly because the means employed by the libertarian paternalists to achieve their goals are not transparent, the accountability deficit of the libertarian paternalistic intervention can be particularly pronounced.

Second, the assumption must be made that the users of choice architecture will always employ this technique in the best interests of the nudged individuals. However, 'government' is not an abstract entity (benevolent or malevolent, according to one's political bent). It is made up of individuals, with their own interests, (first and foremost to be elected and re-elected), who operate through a bureaucracy which is in turn made up of real people with their agendas, interests, biases and, yes, bounded-rationality limitations. When this is taken into account, one of the main criteria about the desirability of a political system is how effectively it allows the monitoring of the actions of the rulers, and the alignment of the interests of the citizens with those of the politicians. The weaker the ability to monitor, the more the electorate has to rely on the benevolence of the ruler. The libertarian paternalistic literature says little on this important

[10] Friedson (2010)

[11] *ibid*, page 99, emphasis in the original.

point, despite a well-developed body of theory having been developed since the 1970s (Public Choice Theory) that deals with this type of problems.

In short: precisely if human beings are indeed so bad at making choices as the libertarian paternalists claim, exploiting these shortcomings, even 'for their own good', can be dangerous – the more so, the more the nudges that work around our cognitive shortcomings are effective. '*The slumber of reason generates monsters*', Goya wrote[12]. Turning these monsters into pleasant dreams without waking up the sleeper may perhaps be possible. But is it safe? Is it desirable? In this paper I argue that it may not be.

2 Justification and Conditions for Effectiveness of Libertarian Paternalism

I assume the reader to be familiar with libertarian paternalism, and nudge, and I point to *Nudge* by Thaler and Sunstein as a useful reference point. However, there is a much wider and more technical literature[13] where the libertarian paternalistic positions are articulated more fully and precisely. For the purpose of my discussion, it is enough to point out that it is through the subtle interplay between the System I ('emotive') and System II ('rational') modes of reasoning that the libertarian paternalists attempt both to steer the behaviour of individuals, and to win their libertarian stripes. In what follows I will explain how they try to accomplish this feat.

2.1 The Rationality of the Libertarian Paternalists

In *Nudge*, Sunstein and Thaler refer to an agent who always acts according to her rational self as an *Econ* (their 'nickname' for *Homo Economicus*). In the libertarian paternalistic literature *Econs* share the standard cognitive and decisional features of *Homo Economicus*: they 'can think like Albert Einstein, store as much memory as IBM's Big Blue, and exercise the willpower of Mahatma Gandhi'[14]. We can also look at Thaler and Sunstein (2003), where the authors first point out that the preferences revealed by actions of individuals do not always bring about the best consequences for the same individuals;[15] as a next

[12] *El sueño de la razon produce monstruos*, inscirption on Plate 43 of *Los Capricios*, 1799

[13] See, eg, Sunstein, Cass R, Thaler, R H, (2004), Sunstein, C, Thaler R, (2003), Thaler R, H, Sunstein, C, (2009), Sunstein, C, (2005), Mullainathan, S, Thaler R, (2001), New, B, Le Grand, J, *(forthcoming, PUP)*. For a legal perspective see, eg, Jolls C, Sunstein, Cass R, (2005), Sunstein, C, (1997), Chapter 8 in particular, Benartzi, S, Thaler, R H, (2002). For specific applications, see, eg, Thaler R H, Benartzi S, (2004). I do not consider in any detail in this note the related idea of asymmetric paternalism – see, however, Camerer et al, (2003). Roughly speaking the main distinction is that asymmetric paternalists propose measures similar to those put forth by the libertarian paternalists, but do not make, or play down, the claims that they do not infringe on personal autnomy.

[14] *Nudge*

[15] '...[libertarian paternalists] do not always equate revealed preferences with welfare...', Thaler and Sunstein, (2*003)*, page 175

step, they point out that libertarian paternalists '...emphasize that individuals sometimes make *inferior* choices, choices, that is, *that they would change if they had complete information, unlimited cognitive abilities and no lack of willpower*...'[16] After claiming[17] that there is little support for the view that individuals always, *or even often*, make choices in their best interest, Thaler and Sunstein (2003) then state that '...[r]esearch by psychologists and economists over the past three decades has raised questions *about the rationality* of the judgements and decisions that individuals make. People do not exhibit rational expectations, fail to make forecasts that are consistent with Bayes' rule, [...], exhibit preference reversals (that is they prefer A to B and B to A) and make different choices depending on the wording of the problem...'[18] In short, it is clear that the rationality libertarian paternalists refer to is the neoclassical version of rationality of *Homo Economicus*.

Now, the rationality of *Homo Economicus* is unforgivingly inflexible. Any deviation from the one accepted way of choosing rationally is for neoclassical economists (and hence for the libertarian paternalists) an error and a cognitive failure. Irrational decisions are 'mistakes'.

Furthermore, the deep-rooted nature of the deviations from these rules negates the possibility that fallible humans may even approximate the way *Econs* reason: departure from Bayes' rule invalidates the ability to make *rational* updates of one's beliefs; and preference reversals and frame-dependence implies that agents cannot 'objectively' rank future uncertain outcomes on the basis of utility maximization[19].

Finally, in the quotes above there is also the implication that the rationality of *Homo Economicus* can never give rise to *ex ante* poor decisions. Indeed, libertarian paternalists never mention any reason why *Econs* may make *ex ante* poor choices.

In sum, according to the libertarian paternalistic view, the Bayesian updating, discounted-expected-utility-maximizing, game-theory-consistent System-II self is the ultimate maker and arbiter of good choices. Conversely, any departure from the neoclassical economical way of reasoning is the source (or, at least, an important source) of poor decision-making; and rational decision-makers do not make *ex ante* poor choices.

This, I think, is clear. I claim, however, that libertarian paternalists do not simply endorse, but they positively *need* the neoclassical version of rationality. I explain why it is so in the next subsection.

[16] *ibid*, page 175, my emphasis.

[17] ibid, page 176

[18] *ibid*, page 176, my emphasis.

[19] The lack of transitivity of choices actually stops in their tracks even the first moves of classical decision theory under certainty. *A fortiori*, the von Neumann expected utility maximization framework does not stand the slightest chance of lifting itself off the ground.

2.2 Why Libertarian Paternalists 'Need' Neoclassical Rationality

Libertarian paternalists want to arrange choices in such a way as to nudge the individual towards the choice that she herself (or, rather, her System-II self) would make, and they want to do so in as 'objective' a manner as possible.[20] But for the hidden preference of the System-II self to be intelligible from the outside (ie, for the libertarian paternalist choice engineer to be able to guess what the individual 'really wants'), it would help immensely if the preference were unambiguous and obtainable by 'objective' and almost mechanical means.

The assumed uniqueness of rational, informed choices then comes to play a very important role in the libertarian paternalistic programme. Indeed, if serious, rational deliberation could lead to materially different conclusions, which one of these conclusions should the libertarian paternalist encourage? The adoption of the neoclassical rationality of *Homo Economicus* therefore greatly helps the libertarian paternalists in guessing what people 'really want'.

Note that two apparently simple and direct solutions ('Why don't we just ask people what they prefer?', or 'Why don't we observe what they choose?') are not available to the libertarian paternalists. 'Observing' would not work because, according to the libertarian paternalists revealed preferences (preferred choices) cannot be relied on to lead to welfare (preferred outcomes).[21] And as for asking people what they like, the libertarian paternalists insist that no way of presenting information is 'neutral', and therefore the observed choices will be strongly influenced by *any* framing.

2.3 Internalities

The concept of *ex*ternalities is well established in economic thinking. It refers to the situation when positive or negative repercussions from an economic activity affect third parties, and these effects are not captured in property rights (are not 'priced in'). I discuss the relevance of the concept of externalities for the libertarian paternalistic programme at length in *Taking Liberties*. What matters for the present discussion is the distinction between the Pigovian and Coasian views of externalities: in the first, the party who suffers from the non-priced externality (say, the house dweller in the airport fly-path) is seen as an 'innocent by-stander'; in the latter, both parties stand to gain or to lose from any arrangement that is chosen to deal with the problem. Any arrangement, from the Coasian perspective, involves costs and benefits. There is not a 'victim' and 'perpetrator'.

[20] Steering an individual towards what her rational self would prefer can be seen as an instance of the informed-preferences view of utility. Note, however, that this view of utility is usually associated with some sort of programme of informed-choice formation, whereby the individual is encouraged, coaxed or trained to reach her informed ('rational') preferences. This 'maieutic' stage in totally by-passed by the libertarian paternalists, who, in a way, take it upon themselves to carry out this exploration.

[21] Thaler and Sunstein, *(2003)*, page 175.

Libertarian paternalists borrow and extend the concept of externalities when they describe the situation where the System-II (rational) self suffers the costs imposed by the choices of the irrational System-I self. As both selves are within the same person, they speak of *in*ternalities.

It is not clear to what extent the term is a suggestive metaphor, or should be taken literally (in economic terms). If the latter, it must be stressed i) that the internalities the libertarian paternalists talk about have a strongly Pigovian flavour, as the System-II self is decidedly seen as the innocent victim of the System-I prevarications[22]; and ii) that libertarian paternalists do not engage with the various, and highly nuanced, possibilities that Coase[23] puts forth to deal with externalities. In particular, they do not address the problem of interpersonal comparison of utilities (that lies at the heart of any solution of externalities, and motivates Coase's market-based, revealed-preferences, solution). I examine this aspect below.

2.3.1 Interpersonal Comparison of Utility with Internalities

When *ex*ternalities are concerned, it is well known that one of the most vexing problems in economics is how to redress them without having to resort to interpersonal comparisons of utility. The libertarian paternalists face the same problem, in spades, when they deal with the concept of *in*ternality. Faced with the problem of having to make a comparison between the utilities of the two selves within the same person, they implicitly choose the most radical solution: as they choose to listen only to the preferences of the rational self, they set the weight of the utility of the System-I self to zero! Yes, indeed, now inter-self comparison of utilities becomes easy, but the problem has been avoided, not solved. The problem of striking a balance between the interests of the two selves, in a nutshell, is solved by the libertarian paternalists by pretending that the conflict does not exist. It has been made to disappear by focussing, again, on the rational self as the origin of good decision-making.

This observation naturally brings me to the broader discussion of how libertarian paternalists know what people truly prefer.

2.4 Interpersonal Intelligibility of Preferences

For the libertarian paternalist to be able to 'nudge' the behaviour of the individual in the direction that 'she herself would have chosen' he must be able to understand her preferences. As libertarian paternalists by and large identify the correct way of deciding with the way of reasoning of *Homo Economicus,* this raises the obvious question: how does *Homo Economicus* forms his choices?

In addition to the cognitive features discussed in Section 2.2, *Homo Economicus* is often also characterized as a (possibly enlightened) purely self-regarding[24]

[22] This observation is clearly made in Whitman (2010).

[23] See the discussion in Section 4.4.3 of *Taking Liberties.*

[24] Following Bowles and Gintis (2006) I use the term 'self-regarding' rather than 'selfish' 'to avoid the circularity arising from the fact that all uncoerced actions are motivated by

agent. In this account *Homo Economicus* only tries to maximize his own welfare, with no concern for the welfare of others.

This second set of attributes (Edgeworth's 'selfishness axiom') are by no means logically necessary, as Binmore (2009) clearly points out[25], and the modern revealed-preferences approach[26] allows economists to escape Edgeworth's modelling strictures that "the first principle of economics is that every agent is actuated only by-self interest."[27,28] There is therefore nothing in modern utility theory that requires the deciding agent to be purely self-regarding.

Let's see what this means for libertarian paternalism. To begin with, it would be grossly unfair to maintain that they try to bring about the choices that the rational psychopathic version of *Homo Economicus* would pursue. A more charitable reading of the position of libertarian paternalists could with little strain interpret their endorsement of the choices made by *Homo Economicus* as follows: *Homo Economicus* suffers from no bounds in his analytical and reasoning abilities, and, in particular, displays consistent preferences; and the goals he would like to attain are not necessarily self-regarding ones.

There is a problem with this interpretation, however. As we just saw, in modern economic theory preferences are usually arrived at via observation of choice behaviour. Revealed-preference theory, however, does not have to contend with the coexistence of two selves, each one with conflicting and often irreconcilable goals. With revealed preferences, 'what we see is what we get'.

Relying on the observation of preferences to deduce what an individual 'really wants' however becomes problematic when two selves are at play. If libertarian paternalists want to 'listen to the rational self', they must have an ability to guess its preferences by filtering out the 'noise' produced by the System-I – a problem with which traditional economists do not have to concern themselves.

Doing so can't be easy. A reasonable thumb-nail sketch of the preferences of real individuals would point to a mixture of self- and other-regarding considerations. When the exact mix of altruism and egoism in each individual is variable and unknown, *and we cannot rely on their revealed preferences to guide our interventions*, guessing what each individual would ultimately really want becomes extremely difficult.[29]

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preferences and hence may confusingly be termed selfish, leaving only those actions that violate one's preference ordering to be called unselfish'.

[25] See Binmore, K (2009), pages 20-21.

[26] Already in the early 1970s, Sen, (1973) was writing: '...the approach of revealed preference has gradually taken hold of choice theory in general and of demand theory in particular...'. Since then the prevalence of the revealed-preference approach has, if anything, increased.

[27] Edgeworth (1881). See also Skidelski, page 11, Chapter 10

[28] Paul Samuelson (1937) argued that via revealed-preference theory one could 'develop a theory of consumers' behaviour freed of any vestigial traces of the utility concept.' However, Sen (1973) argues convincingly that, unless the term 'revealed *preference*' is an 'elaborate pun', a utility- (or, at least, preference-) based interpretation is difficult to escape.

[29] This, by the way, is one explanation of why the unnecessary selfishness axiom is still so popular in traditional economics, and why the self-regarding preferences of *Homo Economicus* are considered 'customary'. Assuming that individuals are motivated purely by self-regarding concerns may not be logically necessary and may well constitute a very crude approximation of real human behaviour, but it does make the modeller's life much easier, by facilitating

This being the case, how can the libertarian paternalists claim to know what the System-II self wants?

One possible way to answer this question is to appeal to a very narrow meaning of rationality, and to couple this narrow definition with the 'selfishness axiom'. But the libertarian paternalists are likely to find unpalatable this way to exit to the impasse. They are therefore caught in a dilemma: either they choose to nudge individual towards the easily-guessable, but often-close-to-sociopathic, choices of the purely self-regarding *Homo Economicus*; or they allow for a more nuanced, and 'nicer' view of humanity, but, to remain *libertarians*, have to divine the exact mix of self-and other-regarding preferences of the System-II self of each individual.

These considerations move from a discussion of what I understand libertarian paternalism to be, to his logical implications. I broaden the scope of this discussion in the next section.

3 Libertarian Paternalism in Its Philosophical Context

After describing – fairly, I hope – the distinctive features of the original parts of the libertarian paternalistic programme, I think it is useful to briefly place it into its wider philosophical context, because the combinations of ideas it brings together may be novel, but its philosophical roots are ancient and illuminating.

3.1 Appetites and Freedom of Choice

There are (at least) two philosophical views about liberty and rationality that have a deep (if implicit) bearing on the libertarian paternalistic outlook.

The first has to do with what makes a decision-maker free. The debate goes all the way back to Plato, who in *Timaeus* speaks 'of the two souls of man, one governed by reason or the spirit and the other by the affections of the flesh'.[30] 'The [...] devices of our hearts must be restrained in the same way tat a 'wylde and kicking horse' must be controlled 'with sharpe spurres' to 'subdue his fierceness' '.[31] It does not take much to read an early version of the System-I and a System-II account of the human mind. But the interesting part is the next logical step: 'A man governed by the affections [System II] [...] cannot be said to be truly free; if his reason 'followeth whether so ever the affection or

(enormously) the task of guessing what a given agent might prefer in a given context. In particular, *assuming a purely self-regarding rational agent, makes the task of interpersonal intelligibility possible.*

[30] Now, the referencing here is fiendlish tricky. The quote is in Skinner (2008), page 28, footnote 41, who quotes Erasmus (1533), (*A booke called in latin Enchiridion militis chrostiani, and in englyshe the manuell of christen knight*), who reports in this book his translation of Plato's *Timaeus*.

[31] ibid, page 28.

appetite calleth', he is living in 'certeyn and assured bondage'[32]. In this view, *we are therefore not free if we follow our (System-I-driven) appetites.*

This view of freedom remained prevalent and orthodox in classical antiquity, and evolves through St Augustine, scholasticism, the Renaissance all the way to Kant. 'To act out of passion, it was widely agreed, is not to act as a free man; such actions are not an expression of true liberty but of mere licence'.[33]

From this perspective, if we want to promote a truly free choice, a true *libertarian* programme should endeavour to bring about the choices that reason [System II] would dictate. Similarly, respecting the choices revealed by the visible preferences of individuals (affected as they are by System-I 'affections') is not really freedom-respecting, as 'such actions are not really an expression of true liberty'. In a similar vein, libertarian paternalists may feel justified in claiming that we are not restricting 'true freedom' if we try to circumvent what the System-I 'appetites' would make us choose.

One might well ask, however, to what extent how one can really 'choose', if reason [System-II] will infallibly show the 'correct' course of action. Can reason perhaps suggest conflicting courses of action?

3.2 The Rationality of Enlightenment

This immediately brings to the fore the second important philosophical heritage of libertarian paternalism, ie, its strong implicit links with ideas of positive freedom. In Grays' account '[t]heories of positive freedom have often assumed that one kind of life [...] is best for all human beings; if people fail to pursue such a life, it is because they are not fully rational [...]. This is the basis of ideas of the [...] rational will, which informs ideas of positive freedom [...]. For these philosophers, true freedom is not acting as you wish but *acting as you would if you were rational*.' [34] (I don't need to spell out the parallels with the libertarian programme here, especially when they say that via choice architecture they would like to bring about the choices that a *rational* individual, freed from the 'distractions' of the System-I appetitive self, would make.)

This, of course, is the 'type of rationality' with which Berlin engaged in an *odi-et-amo* relationships throughout his intellectual life. The danger, in Berlin's view, is that, once we accept that the ultimate good can be discerned by rational means, the step to forcing this rational good upon individuals for their own good becomes worringly small – these are, after all, the choices that *they themselves* would make if only they could listen to their rational System-II selves.

There is, of course, a great difference whether one forces, induces or nudges an individual towards the best rational (System-II) choice, and these differences may qualitatively change the terms of the matter. But it is this 'universalizing' and cosmopolitan view of rationality in itself (not just the means adopted to promote it) that leads Berlin to reject the 'utilitarian paradise' in which the

[32] ibid, page 28.
[33] ibid, page 29
[34] Gray (1996, 2013), page 11, emphasis added

choices of individuals, 'however foolish or desperate' they may be, are manipulated and tampered with 'without [their] knowing what they are doing'[35] so as to bring about the maximum rational utilitarian good. This belief in a rationally-ascertainable good, according to Berlin, is the road to the dictatorship of reason, not to liberalism.

I do not imply for a second that libertarian paternalism is the way to the dictatorship of reason – nor do I assume that Berlin's views on the Enlightenment are the ultimate word on the political consequences of the Enlightenment *Weltanschauung*. However, the Enlightenment view of a universal, disembodied rationality, and the idea that listening to what this rationality tells us truly sets us free, can be discerned in the libertarian paternalistic view of what constitute good decision-making. In particular, this view of rationality can shed light on important parts of the libertarian paternalistic programme, such as the interpersonal intelligibility of preferences. And Berlin's fear that the 'foolish' choices of individuals may be (however gently) manipulated 'without [their] knowing what they are doing' has some resonance when we remember that one distinctive features of the libertarian paternalistic programme is to harness (in their own words, 'exploit') the cognitive limitations of an individual for her own good, while flying below the radar screen of rational deliberations.

There are many more interesting links between libertarian paternalism and the political-philosophical debate, which, for the sake of brevity, I cannot make here.[36] I simply mention the important distinction between instrumental and autonomy-inspired libertarianism. Regarding the latter, libertarian materialism maintain that they respect the 'dignity of autonomous choice' thanks to the easy reversibility of their nudges. As I try to show below, I believe that this claim is overstated. As for the instrumental aspect of liberalism, libertarian paternalists claim that unfettered choices are so vitiated by deep-rooted human cognitive deficiencies, that their outcome is optimal neither for the choosing individual, nor for society; and that so poor are the outcomes produced by these choices that they can be easily improved upon by the unobtrusive interventions they suggest. Paternalism becomes, in their view, not just permissible, but positively called for: we have 'to abandon the less interesting question of whether to be paternalistic or not.'[37]

4 Existing Lines of Criticism

Many different strands of criticism have been levelled at libertarian paternalism. The literature is vast, and the reader is referred to *Taking Liberties* for a lengthy, but still incomplete, discussion. In this paper I simply highlight some main strands.

- Some authors (see, eg, Glaeser (2006), claim that consumers are better incentivized than voters or government officials to make the best choices

[35] Gray (1996, 2013), page 29

[36] The reader may want to look at Section 2.2 and passim of *Taking Liberties*.

[37] Nudge (2008).

– and, therefore, implicitly advocate market *laissez faire*. I explain in Section 4.3 of *Taking Liberties* why I do not find this strand of criticism compelling. Along distinct, but related, lines, some critics express a slipper-slope concern, which I address below.

- Other authors (among whom, again, see Glaeser (2006)), are afraid of putting into the hands of an already-powerful government an additional, powerful and difficult-to-monitor tool of 'persuasion'. I argue in *Taking Liberties* that, paradoxically, hard paternalism, by virtue of its being more visible and 'irksome', may be less dangerous, as the electorate has the 'nuclear' option of voting an over-reaching government out of office.

- Other authors (see, eg, the body of work by Gigerenzer in the Bibliography) argue that human rationality may well be bounded, but that in most cases it can be surprisingly effective. Their 'fast and frugal heuristics' stand in stark contrast to the cognitive blunders regularly perpetrated by the System-I self in the behavioural finance literature. They also claim that the 'faulty' decisions reached by our impulsive self may be faulty in a laboratory setting, but may well have surprisingly adaptive features in real-life situations.

- Some commentators (see. eg, Whitman, 2006) look at the internality concept, point to the Pigovian ('innocent-bystander') view of internalities the libertarian paternalistic take of this construct implies, and point out that, for all their professed 'value neutrality', libertarian paternalists cannot help but 'take sides' between the two selves. See the discussion in *Taking Liberties* in section 4.5

5 The Libertarian Paternalistic Rebuttals and Their Validity

5.1 What Constitutes a Valid Defense?

In this section I sketch the rebuttals by the libertarian paternalists of these criticisms (as presented, for instance, in *Nudge*), and state when and why I do not always find them convincing. I also present some arguments that the libertarian paternalists have not made, but that could strengthen their position.

As for what constitutes a defensible libertarian paternalistic rebuttal, their argument would *not* be helped by arguing that this or that goal they intend to promote is good, either self-evidently, or in their eyes, or in the eyes of the majority of the population, or of the expert. Nor is their position helped by showing that their proposed piece of choice architecture is very effective. They will defend successfully their case if they can show that the course of action towards which they try to nudge an individual is what she 'really wants'; if they manage to give a coherent meaning to the last expression ('what she 'really wants' '); and if they can show that the measures they employ could

de facto (not just theoretically) be rejected at little cost (where 'cost' includes information cost, transaction cost, and various other 'frictions').

The first two requirements ensure that their interventions are paternalistic in the sense that they are in the best interest of the person who has been nudged. The last one that it is libertarian.

5.2 Slippery-Slope Arguments

Thaler and Sunstein (2008) identify in *Nudge* as 'slippery-slope' criticisms the objections of those who fear that a government that starts from libertarian paternalistic measures might end up implementing hard paternalistic ones. Thaler and Sunstein characterize as slippery-slope concerns the worry, for instance, that 'Governments that start with education might end up with stiff fines and even prison terms'.[38]

Mounting a solid defense against slippery-slope arguments is in general not difficult.[39] However, Thaler and Sunstein (2008) do not adopt the standard lines of defense. They complain instead that the slippery-slope objection does not address the question whether the measures they advocate are good or bad: '...[i]f our proposals help people save more, eat better, invest more wisely, and choose better insurance plans and credit cards – in each case only when they want to – isn't that a good thing? If our policies are unwise, then it would be constructive to criticize them directly rather than solely only on the fear of a hypothetical slippery slope...'

This defense is puzzling on two accounts. First, libertarian paternalism is a *method*, not a set of specific policies. When they claim that theirs is the 'true new Third Way', surely libertarian paternalists cannot be referring to their dietary plans, or to their suggestions about the need to save for retirement. Appeal to the desirability of the goals does not justify, in itself, the legitimacy of a paternalistic intervention – soft or hard as it might be.

The second objection is that such an appeal to the goodness of the measure in itself could be made by a benevolent dictator: "Aren't our policies good? If so, why do you complain about our methods?" One of the central tenets of a *liberal* democracy surely must be that the *means* by which a measure is enacted should at the very least be taken into account alongside the desirability of the measure itself.

As a consequence, the line of defence chosen by Thaler and Sunstein does not really address the slippery-slope objection. Their failure to do so can be understood from a perspective that places great the emphasis on the goodness of the outcome, but gives relatively little intrinsic value to act of choosing, either in an absolute sense or as a 'good' to be ranked alongside other desirable goods.

I concur with the libertarian paternalists that the slippery-slope argument *as presented above* (ie, as a transition from libertarian to hard paternalism) is not particularly convincing. However, my own rebuttal of the slippery-slope

[38] Thaler and Sunstein (2008), *Nudge*, page 236

[39] See *TL*, Chapter 5, Section 5.2, for classic lines of defense.

objection rests on the observation that, as paternalistic measures morph from libertarian to hard ones, they become more and more 'visible': changing a default option may easily slip below the radar screen of a person's attentiveness (and, indeed, it is supposed to do so); but sin taxes, 'stiff fines or even prison terms' are progressively less likely to do so. As they become more overt, they allow easier detection and, if disliked, possible rejection by the electorate.[40]

There is, however, a different slippery-slope criticism that I *do* find worrying, and that the libertarian paternalists do not seem to address: some nudges might become progressively more intrusive (and effective), *while remaining below the radar screen of critical appraisal (and sometimes even of consciousness)* – ie, while firmly remaining in the *libertarian* paternalistic camp. For instance, Thaler and Sunstein (2008) write: 'If people want to lose weight, one effective strategy is to put mirrors in the cafeteria. When people see themselves in the mirror, they may eat less if they are chubby. Is that okay? And if mirrors are acceptable, what about mirrors that are intentionally unflattering?...'[41]

Admittedly, Thaler and Sunstein do not answer in the affirmative, but they do not say 'no' either. Their position is so tentative is that they barely reject even subliminal advertising (as long as it is for 'good causes', of course...): '...subliminal advertising does seem to run afoul of the publicity principle[42]. [...] But what if the use of subliminal advertising were disclosed in advance? What if the government openly announces that it will be relying on subliminal advertising in order, for example, to combat violent crime, excessive drinking, and the failure to pay one's taxes? Is disclosure enough? We tend to think that it is not...'[43]

Thaler and Sunstein answer again in the negative, but theirs is hardly a flat-out rejection. And their qualified rejection ('*We tend to think that it is not*') does not address the question of what exactly distinguishes pre-disclosed subliminal advertising from the techniques suggested by the libertarian paternalists. The answer remains unanswered because the libertarian paternalists do not give us a coherent 'demarcation criterion' on the basis of which they can draw the acceptability line.

The slippery slope I am worried about in the libertarian paternalistic programme is therefore the use of progressively-less-visible means of persuasion: the less overt these modes of influencing are, the less an individual is in a position of reversing the 'default option'; the greater the risks in wrongly guessing what the individual really wants (assuming, that is, that we are dealing with an absolutely benevolent choice engineer); the smaller, in general, the degree of monitorability of the whole process. In short, the slippery slope I am worried about does not lead from the reversal of default options to stiff fines and prison sentences, but to subliminal advertising.

[40] 'Hard paternalism in the form of tax rates or bans is easy to monitor and control; soft paternalism is not.', Glaeser, (2006), page 135.

[41] Nudge, page 244

[42] I discuss the publicity principle below. Incidentally, it *obviously* runs afoul of the publicity principle, it doesn't just 'seem to'.

[43] *Nudge*, page 246

5.3 The Preservation of the *Nominal* Freedom of Choice

Most of the criticism of the libertarian paternalistic programme has come from those libertarians who resent infringements of freedom of choice in all shapes or forms. Against these critics, libertarian paternalists present themselves as enhancers of free choice: '...Our proposals are emphatically designed to retain freedom of choice. In many domains [...] we would create such freedom where it does not now exist. *So long as paternalistic intervention can be easily avoided* [...] the risks decried by antipaternalists are modest...'[44]

I find this argument unconvincing. The libertarian paternalists first endeavour to show that *nominal* freedom of choice is vacuous. Let's accept their argument: let's accept, that is, that, in real life, individuals routinely fail to exercise their choices in ways that would be advantageous to them (say, enrolling in a pension), even when the barriers to making the right choice are virtually non-existent.

Consider then, say, the case of a 'bad nudge' (such as the instance of the 'suggested credit card payments' which I discuss at length in Section 3.6 of *Taking Liberties*). Professor Neil Stewart, a psychologist at Nottingham University,[45] has convincingly shown that the usually-very-low 'suggested payment' proposed by credit card companies is an instance of irrelevant anchoring and that it 'nudges' customers towards repaying too small a fraction of their bill (thereby ultimately incurring higher borrowing costs). By doing so, *The Economist* suggests, "in the real world, this would roughly double interest charges".

Now, this is a perfectly reversible nudge, and there are no impediments to repaying a different amount than what suggested. The only difference is that the nudge is designed to 'trick' rather than help its target. Surely, the libertarian paternalists would oppose these choice settings. They would argue that individuals, while nominally free to exercise the 'good' choice, *de facto* fail to do so for a variety of cognitive shortcomings. What is the value of such nominal freedom of choice if it is routinely not exercised even when the cost of doing so is very low, the libertarian paternalists would ask? *But this argument cuts both ways*: what is the autonomy value of the reversibility of the 'good' choices set up by the libertarian paternalists if it is not exercised, ie, if the nudges of the libertarian paternalists are so effective?

Admittedly, there will always be some rare individuals who are so critically minded to be able to reverse the choice engineering deployed by the libertarian paternalists. Probably, however, these rare individuals would easily avail themselves of their freedom of choice in the first place – ie, before any nudge had been implemented. See, in this respect, the very clear discussion in Mitchell (2004).

Surely, libertarian paternalists would welcome, and indeed actively seek, nudges that are *more*, not *less*, efficient. But as the effectiveness of the nudge approaches 100%, real freedom of choice disappears. Consider again the follow-

[44] *Nudge*, emphasis added

[45] Quoted in *The Economist*, December 13th, 2008, page 88, Credit Cards – *A Nudge in the Wrong Direction*

ing striking fact, quoted in Fridson (2010): '...Those who wrestle with political questions of choice and freedom may puzzle over the fact that almost 100 percent of Austrians have agreed to be organ donors but only 12 percent of Germans have done so. The difference is that Germans must opt in, whereas the Austrians are asked whether they do *not* want to become organ donors...'[46]

This dramatic difference in take-up rate should make enthusiastic supporters of easy reversibility pause for thought. It may well be that no choice setting is neutral[47]. But the fact remains that the almost 100 percent observed 'agreement' to being a donor tells us absolutely nothing about how many Austrians 'truly' would want to donate their organs. It only tells us that a very effective nudge has been devised. Autonomy of choice has patently not been preserved, the more so if it is true that no way of presenting information is neutral: '...There is no such thing as a neutral design[48] ...Small and insignificant details can have *major* impacts on people's behaviors. A good rule of thumb is to assume that everything matters...[49] In many cases, some kind of nudge is inevitable, and so it is pointless to ask the government simply to stand aside[50]...'

If this is true, how can the libertarian paternalists claim then, that *their* choice set-up brings out the 'true' preference of the chooser? And how can they maintain that the high acceptance ratios reveal anything about System-II preferences of the individuals?

These two related points are very important, but seems to have been poorly explored in the literature. As a consequence, libertarian paternalists have had a surprisingly 'easy ride', and have rarely been criticized on the grounds of the *effective* reduction in decisional autonomy that even easily reversible nudges entail. The argument therefore deserves more careful discussion, and I explore it immediately below.

5.4 Forcing People to Choose

Some critics accept the libertarian paternalistic claim that, when choices are driven by the System-I self, no design is neutral, but point out that, if that is the case, then one should try to 'force' the rational self to come into play. This could be achieved by inducing the individual to choose.[51] The position of the libertarian paternalists on what they called 'coerced choosing'[52] ranges

[46] Fridson (2010), page 99, emphasis in the orignal. The text speaks of 'agreement'. It would be more correct to speak of 'absence of disagreement'.

[47] This may well be true, but I discuss in the next section that there often are realtively easy ways around the tyranny of framing.

[48] Nudge, page 3

[49] ibid, page 3, my emphasis

[50] ibid, 237

[51] I am aware, of course, that forcing an individual to choose does not eliminate the effect of framing and other biases. However, by requiring a person to 'step back' and make an active and deliberate choice, it is more likely that the reflective self will be brought into play. Libertarian paternalists may be right that no choice setting is neutral, but, surely, some settings may induce a reasoned (the libertarian paternalist would say 'rational') choice better than others. Forcing people to choose may be just one such choice setting.

[52] see,eg, Sunstein (2005), page 186

from qualified acceptance to outright opposition. For instance, Sunstein (2005) correctly points out that: '[c]oerced choosing honors freedom of choice in a certain respect; but it does not appeal to those who would choose not to choose, and indeed it will seem irritating and perhaps unacceptable by their lights...' Sunstein raises two distinct objections here: the first is that forcing to choose is a paternalistic measure in itself; the second is that some (many) individuals would actually strongly prefer not to choose. The first point is valid: forcing to choose is forcing a course of action upon an individual because the outcome is thought to be advantageous for her – and, therefore, it fits even a strict definition of paternalism; I am less convinced by the claim that, for non-trivial matters, people often prefer not to choose. I believe this to be far less common than Sunstein (2005) (and Thaler and Sunstein (2008)) maintain (see my discussion below). However, even if this dislike were pervasive, there would be a very simple way to respect this 'second-order choice': simply to insert the option 'You choose on my behalf'.

Thaler and Sunstein (2008) make a similar point[53]: 'Although nudges are often unavoidable, we enthusiastically agree that required (or strongly encouraged) active choosing is sometimes the right route, and we have no problem with providing information and educational campaigns. But forced choosing is not always best. When the choices are hard and the options are numerous, requiring people to choose by themselves might be preferred and might not lead to the best decisions. Given that people would often choose not to choose, it is hard to see why freedom lovers should compel choice even though people (freely and voluntarily) resists it...'

There are many strands to this argument, best considered one by one. Thaler and Sunstein begin by stating that nudges are often unavoidable. Note that this is not the same as saying that framing is unavoidable. To the extent that nudging is an *active* act of choice engineering, this statement is considerably stronger than saying than there is no neutral way to present information. Saying that nudging is unavoidable blurs the boundaries between presentation of information and active intervention ("Something we must do, anyhow!").

Thaler and Sunstein then go on saying that 'sometimes' they 'enthusiastically' endorse 'required (or strongly encouraged) active choosing'. One would like to understand better when this 'sometimes' applies, as this statement sits rather awkwardly with other important parts of the libertarian paternalistic programme. Consider a few examples. When libertarian paternalists suggest to change the default option about organ donation; when they put forth their Save-More-Tomorrow savings plan, that hinges on people failing to understand the difference between nominal and real income; when they recommend that certain pension plans should become default; when they suggest a placement of healthy food items in the display of a canteen – in all these cases, how exactly are libertarian paternalists 'enthusiastically encouraging' active choosing? Aren't their nudges (their framing) instead based on *preventing* individuals from be confronted with a well-posed question and on relying instead on the pervasiveness

[53] *Nudge*, page 243

of cognitive biases to obtain the desired goals?

The next part of the argument (since 'people would often choose not to choose, it is hard to see why freedom lovers should compel choice even though people (freely and voluntarily) resists it') is the least convincing, and one would like the authors to provide far more compelling evidence than the entertaining but rather flippant example of the diner who entrusts the waiter with the choices of his dinner wine.[54] In most situations, people actually seem to display great pleasure in the act of choice in itself, and resent being deprived of the opportunity to choose for themselves. I would like to point, for instance, to those people who face the discomfort of queuing for hours and sometimes physical danger in order to cast a vote that, individually, will almost certainly have no effect on the outcome of an election. This example, and many other, suggest to me that individuals actually value greatly the act of choosing *in itself*, resent being deprived of choice, and seldom willingly entrust 'experts' to choose on their behalf.

6 New Lines Of Criticism

In this section I look at strands of criticism that are either original (to the best of my knowledge), or have received little attention in the literature.

6.1 Understanding What Other People Like

For the libertarian paternalist to be able to 'nudge' the behaviour of the individual in the direction that 'she herself would have chosen', he must be able to understand her preferences, and to do so better than she (or, rather, her System-I self) would. However, this task clearly becomes very difficult when the exact mix of altruism and egoism in each individual is variable and unknown, *and we cannot rely on their revealed preferences to guide our interventions.*[55]

Of course, if we understand decision-making in a purely instrumental way, then, yes, in many circumstances, making use of rationality can work very well and it is easy to guess what the most effective course of action should be. But if we include in the term 'decision-making' the formulation of a preference, appeal to rationality raises many problems – not least of which is the circularity of the argument, as I show in the next section.

[54] *Nudge*, page 243: '...If we ask the waiter to select a good bottle of wine to go with our dinner, we would not be happy if he says that we should just choose for ourselves...'

[55] This, by the way, is one of the reasons why the unnecessary selfishness axiom is still so popular in traditional economics, and why the self-regarding preferences of *Homo Economicus* are considered 'customary'. Assuming that individuals are motivated purely by self-regarding concerns may not be logically necessary and may well constitute a very crude approximation of real human behaviour. However, it makes the modeller's life much easier, as making this assumption does facilitate (enormously) the task of guessing what a given agent might prefer in a given context. In particular, *assuming a purely self-regarding rational agent, makes the task of interpersonal intelligibility possible.*

6.2 Circularity of the Argument

One of the fundamental tenets of the libertarian paternalistic manifesto is that the rational preferences of the System-II self should be 'listened to'. We should all reason like *Homines Economici*. But why should this be the case?

One explanation is that appeal to rationality is 'forced upon' the libertarian paternalists by the problem of interpersonal intelligibility of preferences – rooted, in turn, in their refusal to make use of revealed preferences and in their two-selves view of the individual. But this is not the explanation offered by the libertarian paternalists. Their justification, instead, goes along the following lines. We should listen to the rational self, they say, because its counsel reflects the conclusions we reach when we think deliberately and we are not 'disturbed' by the interference of our System-I. This seems reasonable enough, but, to see the problems with this justification, let me rephrase their argument as follows.

The libertarian paternalists effectively say: we should listen to what an individual chooses once she has been given ample time to reflect, she is removed from emotional stimuli, and she is allowed to analyze the problem from a rational point of view. Why so? Because, they reply, this is the decision that would be reached by a person who is given ample time to reflect, who is removed from emotional stimuli, and who is allowed to analyze the problem from a rational point of view.

Presented in this manner the argument is clearly circular, but can it really be a fair rephrasing of the libertarian paternalistic position? The circularity of the argument is not immediately apparent because we tend to think that listening to reason is 'obviously' good. However, the exhortation above is logically equivalent to the following: we should listen to an individual's first, unreflective reactions, before she is allowed to analyze the problem from a rational point of view. Why so? Because this is the decision revealed by a person's first, unreflective reaction, uncontaminated by her rational analysis.

We may not like the conclusion, but, logically, nothing distinguishes the two criteria to give preference one 'System' over the other. Note that the argument would not be circular if the libertarian paternalists said, for instance: "We should follow the rational System II because this is what the virtuous person does – or because it will produce the greatest good for the greatest number, or because there is a moral imperative to do so." Some, or all, or none of the above justifications may be implicit in what libertarian paternalists propose, but, in my reading, they are not explicitly invoked as the logical underpinning of their programme.

So, the libertarian paternalists seem to invoke a neutral and 'objective' preference criterion based on rationality, but, when they claim that the preferences of the System-II self should always be listened to, they are actually making a very strong choice (in Whitman's language[56], they resolutely 'take sides'). Of course, they cannot claim that they are simply listening to what the (whole) person really wants, as, by their own description, there is no single person, but

[56] see Whitman (2010)

two selves. They do not justify their *a priori* choice to listen to the *Homo-Economicus*-like 'self' in order to discover what an individual 'really' wants. However, at least since Hume ('*Reason is the slave of passion*'), the claim that rationality is the origin of our preferences and of our good choices has been vigorously challenged.

The objections raised in this section should not be confused with the statement that reason cannot play a role in determining what goals a person should pursue. Indeed, in the 'informed-preferences' view of utility[57] – a view for which the libertarian paternalists are likely to have a lot of sympathy – reason plays a central role in determining which goals one should pursue. Rather, in this section I have made two linked but distinct points. The first is that reason *by itself* cannot uniquely identify a set of preferences. As Sen's says, what are required are 'preferences over preferences'. The unbounded rationality of *Homo Economicus* is of little help in determining the preferences over preferences that are at the root of the informed-preferences approach. The brief allusions to the guiding role played by reason in the Enlightenment programme, and Berlin's objections, presented in Section 3.2 could be profitably revisited at this juncture.

The second point is that sometimes listening to the rational self can be ineffective even from a purely instrumental perspective, ie, even once the preferences over preferences have somehow been established. I turn to this aspect below.

6.3 Instrumental Effectiveness of Rationality: Strategic Interactions

Once a preference has been formulated, the primacy of our System II when it comes to efficiently obtaining it may appear easy to defend. Yet, even in this exquisitely instrumental area matters are not always so clear-cut. In this section I look at those situations (typically found in the domain of strategic interactions), where his infallible rationality does not help, and actually hinders, *Homo Economicus* in reaching instrumentally optimal choices.

There is a long list of strategic games (of course, the Prisoner's Dilemma, but also, for instance, the Tragedy of the Commons, the Diner's Dilemma or, to some extents, the Stag Hunt game) where the Nash equilibrium that selfish, rational players will choose is not only 'bad for society', but also bad for the rationally-choosing individual.[58] I want to argue in this section that for these types of strategic interaction relying on pure rationality may not provide the best instrumental answer *irrespective of whether we deal with a purely self-regarding*[59]*, or with an 'altruistic' System-II self.* Let us start from a selfish

[57] I briefly describe and discuss this view in Section 8.1.

[58] One should not think that these 'dilemmas' are artificial and specially hand-picked situations rarely met in real life. As Sen (1973) points out, just the Prisoner's Dilemma 'has obvious references on the theory of optimum savings, on taxation theory, on allocation decisions involving externalities and public goods, and on a number of related issues'.

[59] As we have seen, this is a reductive view of rationality: nothing requires a rational decision-maker to display purely-self-regarding preferences. However, as pointed out above, making this

Homo Economicus.[60]

Consider selfish individuals engaging in a Prisoner's dilemma. Here rationality doesn't save them from, but condemns them to, sub-optimal outcomes. And this sub-optimality applies both to society, and to the choosing individual. So, even if each individual had purely self-regarding preferences, appeal to pure rationality can be self-defeating, in the sense that listening to the System-II self would not bring about outcomes *that a self-regarding System-II would be happy with*: rationality is the problem, not the solution.

What if the deciding person were motivated by non-self-regarding preferences? Would rationality be an instrumentally good guide in this case? For instance, let us assume that she has a keen sense of fairness, understood as a dislike of income inequality. Then, in, say, the Dictator game, she would certainly not follow the suggestions of a rational player with self-regarding preferences (who would offer 1 cent). But it is not clear what role rationality would play if she wanted to be fair, to guess what the other player may perceive as fair, and, of course, to retain a 'fair fraction' of the dollar. The problem is that in most strategic interactions where players are modelled as being non-purely-self-regarding, it is well-nigh impossible to know what exact mix of self-regarding and selfish motivations the two players will have. The best rational response of rational-but-not-necessarily-self-regarding player A depends on the imperfectly known preferences of the rational-but-not-necessarily-self-regarding player B. Now, in the abstract it is not difficult to translate non-self-regarding strategies in a pay-off table. But, as Hargreaves Heapes and Varoufakis (2005) show, in such a setting the optimal rational response can completely and discontinuously change as a function of the (unknown) mix of altruism and selfishness.

The point here is *not* that introducing more nuanced psychological features in the make-up of *Homo Economicus* makes a rational game-theoretical approach inappropriate: '*Homo Economicus* [...] is apparently unperturbed by accusations of having an inadequate psychology and no capacity to overcome the Prisoner's Dilemma cooperatively'[61]. Instrumental rationality has no difficulty in finding an optimal solution *when the preferences (selfish or otherwise) of the players are perfectly specified*. It is, however, this perfect specification that becomes difficult when we abandon the safe but dismal shores of self-regarding preferences.

In sum: I have argued so far that whenever strategic interactions are at play, it is virtually impossible to state in the abstract whether employing the strict rationality of *Homo Economicus* will always produce the best outcome. And this is true for each player and for society. Therefore the libertarian paternalists

assumption greatly facilitates the guessing of what a person really wants. See the discussion below.

[60] To be fair, libertarian paternalists do not explain carefully what they mean by *Homo Economicus* (*Econ*, in their language) – in particular, whether they endow this agent with purely self-regarding preferences. If they don't, the difficulties they face in justifying why rationality should be the best instrument to reach their goals are, of course, even larger. But, even if they deal with a purely self-regarding agent, reason may not always be the best guide to her actions.

[61] *ibid*, page 269

appeal to the choices of the System-II self to nudge individuals faced when strategic interactions are at play becomes difficult to justify.

6.4 What Is So Rational About *Rational* Choice Theory?

I have discussed at length how important *rational* decision-making is for libertarian paternalists. In particular, they say, if we want humans to behave like *Econs*, they should follow the percept of modern choice theory. (See, eg, Kreps (1988).)

The latter is usually regarded as a paragon of rationality. However, for the theory to have any prescriptive or descriptive power, behavioural *preferences* (which have nothing to do with rationality) must be injected into the 'empty vessel' of the axioms of choice.[62] Two such preferences are the preference for a certain consumption to a non-zero variance lottery with the same expected payoff (this is the basis of risk aversion); and the preference for more to less. It *would* be possible to build choice theory without invoking these two 'behavioural observations', but its descriptive power would be minimal, and its prescriptions close to vacuous.

It should also be noted that some of the building blocks of modern choice theory, such as the practice of discounting future consumption, are deemed perfectly rational in economic theory, but decried as examples of irrationality in philosophical discourse.[63]

I deal with the rationality of choice theory at some length in *Taking Liberties*, but for the present discussion the important point is that we are, of course, at liberty to apply whatever (well-defined) label we wish to anything – and 'rational' could be one such label. However, when strong judgemental and value-laden dimensions attach to these labels, the choice of the 'name of the label' ceases to be neutral: for instance, it ceases to be neutral, for instance, when it 'authorizes' interventions when action are seem as departing from what the 'rationality labels' says.

[62] Of course, axioms are neither rational nor irrational: they are simply consistent or inconsistent.

[63] Persson (2010), for instance, points out that '...it is a well-known fact that, in appraising values located at different points in time, we display various biases, for example, we are spontaneously inclined *to be biased towards the near future*, and to prefer a closer, smaller good to a more distant, greater good. Yet prima facie it seems cognitively irrational to regard such differences purely in timing as evaluatevely significant. This impression is indeed borne out, but not by there being any underlying belief about temporal facts that philosophical analysis reveals to be cognitively irrational...' Persson (2010), page 8, emphasis in the original. See also *ibid*, Part III, Chapters 14 to 17. In a similar vein, Persson (2010) writes: '...Spontaneously, one is strongly disposed to be *biased towards oneself*, that is, one is more anxious to see to it that a desire by fulfilled if it is *one's own* rather than somebody else's. [...] [A]n analysis of the concept of our identity through time reveals this difference to be without rational importance and, hence, this bias to be cognitively irrational...' page 9, emphasis in the original. See also *ibid*, Part IV, Chapters 18 and 29 in particular.

7 Preferences, Mental States and Revealed Utility

In this section I bring together several strands of criticism which have in common an emphasis – neglected, I think, in libertarian paternalism – on the *full* decisional process, (eg, on the associated mental states, on the preferences informing the act of choice, etc), rather than on the *outcome* of a choice act.

7.1 Preferences and Utility

Libertarian paternalists certainly believe in utility maximization, as they set *Homo Economicus* as their gold standard of decision-making. In most applied situations, the concept of utility is currently used in economics in an *as-if* sense, and in this sense asking the precise meaning of utility is of relatively little importance – and perhaps indicates muddled thinking[64]. However, the moment libertarian paternalists claim that they do not want to employ the revealed-preferences window into preferences, then the question of what they *really* mean by utility, and what they want to maximize, does matter.

I argue in this section that the libertarian paternalistic position implies the endorsement of a particular, and far from uncontroversial, view of utility. Libertarian paternalists, for instance, do not seem to entertain the possibility that, say, a person may *rationally* want to be slim and yet refuse to go on a diet. Yet, she may *rationally* hold both preferences because other elements may be at play in the full utility calculus of the choice process that are not visible if one just considers the start and end points (eating generous portions of food and being slim). Indeed, for many individuals going about their lives consistently listening to what their rational selves suggest entails psychological costs they would not be prepared to pay. It is not that these people would not, in the abstract, prefer to be slimmer, to enjoy later in their lives a comfortable retirement, or to live longer. Nor do they doubt that the course of action suggested by rational analysis to pursue these goals would not be effective. Many individuals may be simply not prepared to sacrifice to the attainment of these goals the spontaneity, immediacy and care-free attitude to life that they cherish very highly. Their failure to follow the 'rational' course of action is not due in these cases to a lack of resolve, hyperbolic discounting or other alleged cognitive deficiencies. These people simply include in their (totally rational) calculus of happiness *the process of choice in its entirety* (not only the starting and final points of the process), and the mental states associated with following any course of actions (say, spontaneity, rebelliousness, etc). Coining a term I will explain immediately below, these people are not visible-end-points consequentialists.

The libertarian paternalists, I claim, focus their attention on the observable starting and end points of a choice process, and pay little attention to what 'goes on in between'. This, I believe, can be explained by looking at their implicit interpretation of what utility is, and of what people maximize when

[64] See Binmore (2009) in this respect.

they maximize utility. To explain what I mean, it is useful to take a brief look at the meanings given in political philosophy to the term 'utility'.

One can recognize at least four distinct interpretative strands of utility.[65] The simplest is the welfare-hedonistic interpretation (utility = happiness or, more precisely, utility = experience or sensation of pleasure). The argument presented by Nozick (1974) with his 'experience machine' (if we had a neuro-machine capable of producing constant happiness, would we really accept to give up our 'real' life, and be hooked to the machine for the rest of our lives?) is the best known line of criticism to this welfare-hedonistic interpretation. (By the way, the answer is supposed to be, 'no')

As an attempt to get around the restrictiveness of this hedonistic view, non-hedonistic mental-state utility was introduced. Non-hedonistic mental-state utility is equated with all the experiences that we value in life, irrespective of whether these bring pleasure or not. After all, human beings are observed willingly to engage in activities (running marathons, writing papers on libertarian paternalism, reading Joyce's *Ulysses*) that they seem to highly prize, yet whose immediate effects cannot be described as a feeling of pleasure. However, no matter how complex the mental states associated with these activities are, and no matter how different from straightforward pleasure, Nozick's experience machine still creates a problem for non-hedonistic mental-state utility: '...[s]uperduper neuropsychologists could stimulate your brain so that you would think and feel you were writing a great novel, or making a friend, or reading an interesting book. All the time you would be floating in a tank, with electrodes attached to your brain. Should you plug into this machine for life, pre-programming your life's experiences?'[66]

If your answer is 'no', then the preference-satisfaction interpretation of utility has to be broadened. One way to do this is by saying that "increasing people's utility means satisfying their preferences, *whatever they are*"[67] – which would include, among other things, *not* being hooked up the superduper Nozick's machine if a person did not want to be so connected. This is an interpretation that can be made close to the one implicitly adopted in mainstream economic analysis (revealed-preferences theory). There are several problems, however.

The first is that, for all their limitations, the welfare-hedonistic and the non-hedonistic mental-state interpretations of utility were at least trying to tell us what utility *was*; the preference-satisfaction interpretation is more hollow of descriptive content, and can 'smell of' tautology.

The second problem is that what people prefer may be influenced in a 'constraining' manner by the circumstances they find themselves in. Poverty, for instance, or lack of education can be regarded as capability-limiting conditions[68].

[65] This neat distinction of four strands of interpretation of utility draws on the discussion in Kymblica, (2002), page 10 and *passim*.

[66] Nozick, R, (1974 - 2009), Chapter 3, 42-45

[67] Kymblica, (2002), page 14, my emphasis.

[68] To my knowledge, the capability approach was first advanced by Sen in his Tanner lecture in 1980 (Sen (1980)). For a discussion of poverty as capability deprivation, see Sen (1999a). For the subsequent developements and enrichments by Sen of the concept see the references

People in conditions of physical, economic or intellectual deprivation may also 'adapt' their preferences to 'being happy for small mercies'. In both cases, even if we could guess what, *in their present conditions*, the individual prefers, these adaptive or capability-constrained preferences should not be automatically regarded as the 'true' preferences of an individual. Without going into the merit of the capability and adaptiveness debate, it is important to stress that these considerations can lead to the informed-preferences view of utility, discussed below – a view which the libertarian paternalists do not 'officially endorse', but for which they are likely to have a lot of sympathy.

A third problem with saying that "increasing people's utility means satisfying their preferences, *whatever they are*", is related to how we can gain access to what the preferences of an individual are. A simple solution to this problem (the one commonly employed in economic analysis) is simply to point to what people actually choose when left free to decide. Doing so, however, further widens the gulf between utility and psychological pleasure, and brings about a psychological hollowing out of the concept of utility: '[i]n modern use of 'utility' in contemporary choice theory, its identification with pleasure or desire-fulfillment has been largely abandoned in favour of seeing utility simply as a numerical representation of a person's choice'.[69]

To get closer to a view of utility that libertarian paternalists might feel comfortable with we therefore have to look elsewhere, namely at the informed-preferences view of utility. According to this view, utility is increased not by the satisfaction of *any* preference, but only of the 'rational' or 'informed' preferences. I will say more about this later, but we should not underestimate the difficulty of the task faced by an external observer – and, more to the point, a policy maker – who wants to increase this 'type of' utility and has to distinguish informed and rational from hasty and irrational preferences.

Of course, just observing what the individual chooses is no longer a viable option, as the preference that is satisfied by the observed action may be the uniformed or irrational one. So, how do we know whether the fat person indulging in a double chocolate and vanilla ice-cream has carried out a well-reasoned trade-off analysis between the pleasures of the ice-cream and of a *carpe diem* way of life, and the increased risk of coronary disease, or is ignorant about the links, or has 'adapted' her preferences to her circumstances, or is just 'weak-willed'?

It must be stressed again that, as soon as libertarian paternalists claim that the actions of unfettered choosing agents do not necessarily reflect their 'true' preferences, they bring upon themselves two problems: first, to explain what the hidden utility they want to maximize really is; second, how they can gain access to this hidden quantity.

The latter problem may be tackled in a variety of ways: for instance, by introspection, or by appeal to the universality of rational choices (I have expressed my reservations about this approach in the previous section). However, there is a way to circumvent what is, after all, the problem of interpersonal intelli-

quoted in Quzilbash (2011), page 23. For a critique, see, eg, Sugden (2006).

[69] Sen (1999a), Chapter 3, page 67

gibility of preferences reappearing under a slightly different guise: rather than guessing what an informed person would truly and rationally prefer, a social planner could try to favour those social conditions that facilitate the formation of an informed and well-considered choice, and then let individuals get on with freely choosing. So, for instance, '[w]e may not be able to identify which specific preferences are distorted by false beliefs or adaptive[70] preferences but we can examine the social or cultural conditions under which people form and revise their preferences, to make sure that people have access to appropriate information, and/or opportunities to test alternative ways of life, and/or protection from distorting images or propaganda....' [one could] 'deal with the problem of false or adaptive preferences, not by directly filtering them out, but by eliminating the background conditions that generate such preferences...'[71]

This is an appealing option, in that it makes an effort to respect the decisional autonomy and, in a sense, the human dignity of the choosers. What is relevant for the purpose of the present discussion is that there are two aspects to this enabling and filtering programme. The first relates to the provision of ample and unbiased factual information provision, on the basis of which informed decisions can be made.[72] The second aspect of the enabling and filtering programme is the goal to help decision-makers in reaching not just factually-informed decisions, but also *thoughtful* decisions, decisions, that is, that the individual would not regret after careful consideration.

The libertarian paternalists do not go down this route with great enthusiasm. Their programme is essentially one of context manipulation, not one of fostering of rationality. (Recall that – for our own good – libertarian paternalists want to '*exploit*' and '*manipulate*', not overcome, our cognitive deficiencies.) [73]

7.2 The Libertarian Paternalistic Predicament

The libertarian paternalists therefore find themselves in a singularly difficult position. To begin with, they consider a perfectly-informed and perfectly rational *Homo Economicus* as the perfect decision maker. Despite this, they do not seem to very interested in those measures (the enabling and filtering programme alluded to above) that could help real-life individuals approximate these standards of excellence in decision-making. Instead they seem to take the cognitive

[70] adaptive does not have an evolutionary meaning in this context

[71] Kymblica, (2002), page 19

[72] The libertarian paternalism claim to be enthusiastic proponents of information provision. It remains unexplained, however, how their cognitively impaired decision-maker suddenly becomes able to process the astonishing amount of information they propose to provide.

[73] See, for instance, the following quote from Sunstein (1997): 'in environmental regulation, it is possible to *manipulate* the reference point by insisting that policymakers are trying to "restore" water or air quality to its state at date X; a proposal to "improve" air or water quality from date Y may "code" quite differently. The restoration time matters a great deal to people's judgement.', quoted in Saint Paul (2011), page 84, emphasis added. As Saint-Paul (2011) points out '[t]his means that the way the regulation is presented will have an impact on its support in the population, and therefore on whether the regulation will go through'. Not surprisingly, the word used by Saint-Paul to describe these proposed practices is 'propaganda'.

limitations of humans as biologically hard-wired and practically unavoidable, and resolutely embark on a path of context manipulation and, in general, of exploitation of the boundedness of human rationality. Given the supposed pervasiveness and inevitability of these cognitively limitations, they then refuse to make use of revealed preferences. They are therefore faced with an enormous problem of interpersonal intelligibility of preferences.

And this is where they have to make a conceptual leap, and implicitly embrace a very restrictive version of utility. As their distrust of revealed preferences makes the mental states of individuals virtually inaccessible, and as informed-preference satisfaction cannot be left to the decision maker, they attempt to estimate the utility attaching to the course of action that connect what I called above the *externally visible* start and end points of a decision-making process (the eating of the cake, and increasing the risk of coronary disease).

What libertarian paternalists do not want to, *and cannot*, take into account is, for instance, that the way of life associated with taking constant and conscious care of one's well-being may for some be disagreeable; that others may have a desire to 'rebel against' information 'pushed down their throat', even if with the best intentions; that many individuals may perhaps be as cognitively limited as the libertarian paternalists maintain, but that they have a keen sense to detect manipulation and interference with their decision making. All these 'hidden' aspects of a decision process can play an important role in the overall utility calculus attached to a certain course of action.

All that is available to the libertarian paternalists is therefore a simplified version of consequentialism, where the mental-state consequences of a course of action must be taken out of the consequential calculus.

This, I believe, is a central feature of the libertarian paternalistic programme, a feature that I therefore intend to explore more fully in the following.

7.3 Justifications from the 'Badness' of Biases

As I was queuing at the local corner shop a few days ago, I could not help noticing the purchases of the rather down-at-heels customer in front of me: two bottles of beer, a chocolate snack, a packet of cigarettes and a packet of crisps. The total for this bundle was £8.25 – a price, incidentally, for which a single person can certainly buy a very decent and 'healthy' meal. Now, the libertarian paternalists would probably like to nudge the consumer of the corner shop towards 'objectively healthier' choices. On what grounds, however? The problem here is that the customer did not seem to be acting in any way like 'a giddy and impulsive teenager', succumbing to impulsive choices when he had actually walked into the shop to buy an orange and a bottle of Evian mineral water. As a matter of fact, he actually had to *ask for* the salt-and-vinegar crisps, which were not on obvious display.

Now, we may feel that it would be more desirable if the customer of the corner shop had spent his probably rather limited resources on something 'better' for himself. However, invoking impulsiveness, framing, salience or any of the other related cognitive labels does not seem to come even close to accounting

for his choices. I have no proof of this, of course, but my guess would be that the customer walked into the shop determined to buy exactly two bottles of beer, a chocolate snack, a packet of cigarettes and a packet of crisps. If this is correct, and *if* we think that something should be done to alter these choices, the justification for our intervention can hardly be found in the impulsive and non-reflective nature of the decision-making process (the 'giddy and impulsive teenager' picture of System-I self the libertarian paternalists endorse). Equating choices that we perceive as 'poor' with impulsive (Sistem-I) choices – *with choices that we therefore feel authorized to interfere with* – is an oversimplification. And, given the nature of the libertarian paternalistic programme, not an innocuous one.

If interfering with the choices of the man in the corner shop were deemed desirable, perhaps a social engineer should look at the process whereby these preferences are formed in the first place, ask himself whether it is possible to alter this preference-formation process, and how this interference can be justified. In my view, looking at this trivial instance through the combined lenses of measures against 'adaptation' and of Sen's capabilities approach[74] appears far more productive.

The man in-the-corner-shop is not an isolated example. From organ donation to the choice of medical coverage or of a pension plan, impulsiveness is infrequently at the root of the poor decision-making. The passive acceptance of the default (a form of framing) probably comes closer to a correct description. The only aspect, however, that acceptance of a default has in common with the hasty picking up of a Mars bar from the temptation counter at a supermarket check-out is that they are both labelled as cognitive biases. And if all cognitive biases are bad, then there always is at least a justification for some intervention to overcome them. The 'badness' of cognitive biases (and the assertion that *Homo Economicus* should always trump *Homo Heuristicus*) gives the choice architect (and the libertarian paternalist) a virtual *carte blanche* for intervention, without the need to explore whether other routes (from, say, coerced choice to overcoming the default bias), could be more effective. And even if we believe that the 'satisficing' choice of the man in the corner shop should be interfered with (ie, that his heuristics are 'bad'), the justifications adduced for by the libertarian paternalists are still unconvincing. There is little justification in dismissing 'poor choosers' (if poor choosers they are) as 'giddy and impulsive teenagers', and in proceeding on this basis to our interventions.

This matters, because pointing to the wrong causes of what we may want to change can at best justify ineffective interventions that; at worst it can legitimize rather debatable courses of actions. Few would disagree with, or find much danger in, the gentle steering of a 'giddy and impulsive teenager'. But what if the nudges were not as gentle (read, *effectively* reversible) as originally supposed? And what if the poor choices were not due to impulsiveness and lack of self-restraint?

This discussion leads me naturally to the topic of autonomy of choice, which

[74] See, eg, Sen (1980) and the references in Quizilbash (2011).

I deal with in the next subsection.

7.4 Latent Mental States: The Value of Autonomy

In keeping with the consequentialist focus of instrumental libertarians, libertarian paternalism gives relatively little consideration to the importance an individual may attach to 'making her own mistakes'. When libertarian paternalists do deal with decisional autonomy, their treatment of the topic and, one might say, their very understanding of the nature of the issue can be puzzling. Consider, for instance, an article by Benartzi and Thaler (2002) who claim to address the question of the value of autonomy. More precisely, they set out to discover to what extent, in a particular investment setting, choice agents value autonomy. There is little doubt that, in this piece of work, autonomy is indeed their focus, as the title of their article is *How Much Is Investor Autonomy Worth?* The reader is referred to their paper for methodological details, but the Authors' views about decisional autonomy are clearly reflected in the final words of their paper: '...[d]efined-contribution savings plans, and individual savings accounts within a social security type system, are said to have many virtues, such as [...] the ability to construct a portfolio to match one's tastes. In this paper, we have attempted to quantify the value of this latter feature. Do participants actually gain much in utility by being able to choose their own portfolio? We find that they do not. *Most of our participants find the portfolio of the median participant more attractive than the one they have chosen for themselves*, and this was even true for those who rejected a portfolio customized for them by experts...'[75]

It pays to look carefully at the conclusions reached by Benartzi and Thaler. The conclusions of the paper only indirectly, if at all, address the value of autonomy. The Authors simply draw attention to the fact that investors were happier with the median portfolio than with the one they had themselves chosen. The supposed advantage conferred by autonomy appears to be the ability 'to construct a portfolio to match one's tastes'. But this is only one aspect of autonomy. If a clairvoyant were able perfectly to divine the preferred portfolio of each individual, and arrange it accordingly, the resulting portfolio would, by construction, be built to the individual's tastes. But this has little to do with autonomy.

In particular, this is not the same as asking if the same investors felt better about the whole choice process (ie, the outcome, and how to get there) when they were given little choice. What the paper *does* show is that the portfolio built by the participants was not associated with greater utility than the portfolio of the median investor. The two combined outcomes that would have to be compared in order to say something about the value of autonomy are

1. having a more desirable portfolio, but being denied any choice in composing it (to put it bluntly, having being treated as an only partially competent child when it came to its formation); or

[75] Benartzi, Thaler, (2002), my emphasis

2. having a less desirable portfolio but exercising full autonomy in determining its composition.

The Authors do not really make this comparison – the only comparison, in my view, that can truly tell us something interesting about 'how much investors value autonomy'– because they only focus on the desirability of the resulting portfolio. Using the term introduced above, Benartzi and Thaler are 'visible-end-points consequentialists'.

Admittedly, when the matter of choice is perceived to be of a 'technical' nature, individuals may have little pleasure in exercising their own personal choices. But it should not be taken for granted that also in other contexts (such as choosing whether to smoke or what to eat) the *outcome* of the choice process will always be all that matters. Most people like, at least to some extent, to 'make their own mistakes' and 'to be treated as adults'. For some people, autonomy understood in this sense is just what makes us human.

Failing to make (or rarely making) this important distinction between the outcome in itself and the full choice process (outcome plus the ability or otherwise of choosing) is at the root of the widespread absence in the libertarian paternalistic literature of a sympathetic treatment of autonomy. The same 'impatience' with the issue of autonomy transpires from Thaler and Sunstein's responses to their critics, to which they devote a full chapter in *Nudge*. In a twenty-two page chapter, they devote two svelte paragraphs (approximately) to this issue. After their protestations of being all for autonomy (they say that they 'heartily agree' with the idea that 'people have the right to be wrong, and it is sometime helpful for us to make mistakes, since that is how we learn'[76]), they try to show that they value autonomy by two routes: the first, by pointing to the opt-out rights they embed in their measures; the second, by extolling the virtues of disclosure.

As for the disclosure, I readily concede that it fully respects autonomy. However, there is nothing paternalistic (soft or hard) in disclosure or provision of information. Depending on the application, it may or may not be effective, but it does not present any interesting new development in decision theory or practice. Again, better disclosure can hardly be touted as the True Third Way.

Regarding the opt-out rights, I have made the distinction between real and nominal freedom to undertake a certain course of action, or, specifically, to make a certain choice. For the present discussion, it is enough to point out that the libertarian paternalists go to considerable lengths to show that our System-I self often lets itself be 'tricked' by irrelevant framing – if so, it will also be 'tricked' by the framing chosen by the choice architect. So, if the nudges of the libertarian paternalists – such as changing the default option – are effective, and exploit the decisional inertia of the choosers, then it makes little difference that there is a nominal right to opt out. And if their nudge is *very* effective, then having the nominal right to reverse the nudge makes *very* little difference.

[76] *Nudge*, page 241 – once again, note the instrumental view of decisional mistakes (as a way for us to learn to make the 'right' decisions), rather than an acknowledgement that it is the compound [decision + outcome] that matters.

One more observation about the half page of *Nudge* where Thaler and Sunstein try to deal with the issue of autonomy. The Authors present an example that is both irrelevant to the discussion of libertarian paternalism, and probably consciously chosen to pull at our emotional (very-System-I) chords: '...[H]ow much learning is good for people? We do not believe that children should learn the dangers of swimming pools by falling in and hoping for the best...' Once again, as in the case of the *children's* cafeteria, even most dyed-in-the-wool libertarians acknowledge that, when it comes to children, some degree of paternalism is not only admissible, but probably necessary. The example therefore does not say anything at all about autonomy and libertarian paternalism.

Finally, I want to stress that it is not necessary to subscribe to a view of autonomy as a supreme and inalienable good for the individual to find the libertarian paternalistic treatment unsatisfactory. It only requires acknowledging that individuals draw *some* intrinsic (as opposed to instrumental) pleasure in making decisions about courses of actions that affect their own welfare. If this is the case, and to the extent that libertarian paternalists want to help to maximize the 'true preferences' of an individual, the ultimate absence of the autonomy term in their calculus of her utility maximization is puzzling. This is yet another aspect of the visible-end-points consequentialism discussed above.

7.5 Reversibility of Choices Again

I have repeatedly expressed my skepticism about the claim put forth by the libertarian paternalists that the easy reversibility of their nudges preserves the freedom of choice and the autonomy of the individual. This claim is, of course, central to the validity of their libertarian credentials. Let me briefly summarize the objections I have raised. My points have a lot in common with the criticism raised by Mitchell (2004).

First, recall that libertarian paternalist greatly emphasize the importance of framing; that they claim that *no* decisional setting is neutral; that they are reluctant and very qualified endorsers of coerced choice; and that, as I discuss in the following section, they devote little attention to overcoming cognitive and decisional biases. (See also Mitchell (2004) on this important point.)

As a result of all of this they are perfectly conscious that how they will present choices (the choice-architecture part of their programme) will have a very strong effect on their acceptance or rejection.

They claim, however, that they respect autonomy of choice because their nudges are easily reversible.

However, it would be more precise to state that their suggested choice *could* be easy to reverse by an individual encumbered by the very cognitive limitations that the libertarian paternalists are so keen to point out (and exploit). To the extent that individuals pervasively suffer from cognitive biases, this nominal ability to reverse a nudge will in reality amount to very little. And, perversely, the only (*Homo-Economicus*-like) individuals who are little affected by irrelevant frames and who can therefore easily reverse the libertarian paternalistic nudge are those who could unshackle themselves from irrelevant framing in the

first place – those, that is, for whom libertarian paternalism is, on its own terms, not needed.[77]

In a way, this easy-reversibility of the libertarian argument is vitiated by the presumed validity of their starting assumption: if framing is so important, and if no framing is neutral, then the libertarian paternalists are correct in claiming that nominal ability to exercise a choice is vacuous. But surely this must apply also to the nudges that are put forth by the libertarian paternalistic choice engineer. Both the Germans and the Austrians who display such a dramatic difference in acceptance ratio for organ donation have been 'framed'. Both Austrian and Germans had *nominal* freedom of choice, but neither Germans nor their neighbours have exercised *effective* freedom of choice. From a libertarian paternalistic perspective, they are not any more free to reveal their 'true' preferences in one default setting than in another. Just as elections with 99.9% acceptance for a candidate tell us more about the quality of the democratic process in place than about the virtues of the elected candidate or party, so choices accepted by 'almost 100 per cent' of the population tell us more about the skill and ingenuity of the choice architect than about the true preferences of the decision-maker.

8 Are Our Cognitive Limitations Hard-Wired?

Let me move to another strand of criticism. One of the unstated, but central, assumptions of the libertarian paternalistic programme is that the cognitive biases and inadequacies of an individual are deeply rooted in her psyche – so much so that they tend to be treated as an unchangeable biological given. We are, according to the libertarian paternalists, hard-wired to be imperfect thinkers.

From a libertarian paternalistic point of view the idea that the cognitive limitations of individuals are both widespread and difficult to eradicate is very important: if only the decision-maker were consistently rational, the libertarian paternalists claim, we would be happy to stand aside and to dispense with any intervention in favour of a purely libertarian approach to decision-making; but she is not; hence we must intervene.

In order to justify libertarian paternalistic intervention along these lines, what really matters is not so much the exact extent of human cognitive deficiencies, but whether they are immutable and biologically hard-wired, or whether training, and 'learning how to think' can make a big difference. If it did, the libertarian paternalists would have to explain why they prefer 'nudging' to 'teaching how to think', or, at least, to favouring those choice settings that bring to the fore the rational faculties of the System-II self.

So, in this section I ask the questions: it is true that our cognitive shortcomings cannot be significantly altered by training? And if they could, what would

[77] See again Mitchell (2004) for a closely related argument. Mitchell also points to the attending redistributive effect of libertarian paternalism, with a transfer of wealth from the rational to the irrational. I do not pursue this point, but refer the reader to Section 4 of his work for the very intersting discussion of this point.

this imply for the libertarian paternalistic programme?

8.1 Debiasing and Contextualization

To give concreteness to the discussion, let's consider one of the most frequently cited cognitive biases, ie, influence by an irrelevant frame. [78] This bias seems to be pervasive and important. However, Sunstein and Thaler also point out the settings under which individuals are observed to make cognitively better choices: for instance, when choices are made frequently; when they are less emotionally charged; when enough reflection time (a 'cool-down period') is allowed for the rational System-II self to assert its voice; when the choice is made among a small number of options; when the chooser is an expert in the subject at hand; etc. As Mitchell (2004) points out what seems to be missing is therefore a coherent theory of when framing effects matter and when they don't (or matter little). Absent such a theory, 'we lose information that might educate us on a direct approach to improving decision-making, by which we might foster individual freedom of choice rather than have the central planner make choices for people.'[79]

Sunstein and Thaler also pay little attention to existing empirical work on the decisional settings that make people less susceptible to choice manipulation (debiasing techniques). Mitchell (2004) presents many instances of how contextualization or cognitive problems can be overcome. Further suggestions are presented, for instance, in Gigerenzer and Edwards (2003), who look at the well-know and vexed problem of eliciting conditional probabilities – an area were cognitive biases are supposed to have a very strong impact, especially when the base frequencies are low.)

All of this is encouraging, but the (de)-contextualization route is not danger-free. Presenting information in such a way as to aid the engagement of the rational decisional faculties of the citizen relies on a benevolent social planner, who has at heart the quality of the decision-making process of the electorate. But why should a social planner be so motivated? I discuss this important point in the next section. For the moment, it is important to stress that the 'contextualization game' need not be played in a disinterested manner.[80]

8.2 Moving Beyond Contextualization

If contextualization is no panacea, (and is actually open to the same type of reliance-on-a-benevolent-planner criticism that I level at the libertarian paternalistic programme) are there more palatable solutions? Indeed there are, and they are based on shifting the emphasis away from the *passive* reception of

[78] '...Sunstein and Thaler paint a picture that highlights the pervasiveness of choice framing effects: they write that "in many domains" what people choose "is strongly influenced by details of the context in which they make their choice," and tell us that experience, expertise, and incentives cannot be counted on to protect decision-makers from the influence of the choice frame...'. Mitchell (2004), pags 7 and 8, citations omitted

[79] *ibid*, page 9

[80] See in this respect Gigerenzer and Edwards (2003) and Gigerenzer (2002).

a frame (which could be cognitively well- or ill-designed and which is subject to benevolent or 'malevolent' manipulation), towards the *active* critical examination of a problem by the decision maker. This is, of course, a much more challenging task, and it is not surprising that the majority of the cognitive (as opposed to contextual) debiasing measures reported in the literature[81] appear to be focussed on improving the quality of decision-making in each individual choice instance, by addressing the specifics of each decision-making context. They do not address the issue of how, and if, individuals can be trained to be 'better thinkers' in all situations of life – as opposed to, in the moment of one particular type of choice. But can an individual be trained by be an overall, 'multi-purpose' better thinker?

In this respect, studies such as those by Professor Nisbett[82] are providing important evidence that rationality and intelligence are much more 'teachable' than it is usually thought. What is most surprising is that the improvements that can be achieved via training reach areas of cognitions that were traditionally regarded as innate and, as such, close-to-immutable. Indeed, Prof Nisbett's studies convincingly show that even a cognitive feature such as one's IQ – which had originally been devised exactly to uncover the innate, 'intrinsic' and 'unlearnable' components of an individual's cognitive abilities – can be significantly improved by training. So, if even IQ scores can be improved upon by training, one could hope that the scope for widespread improvement in reasoning in a variety of areas could be substantial.

If it is indeed true that it is possible to improve one's ability to reason – and so, according to the libertarian paternalists, to choose well – libertarian paternalists should not object to devoting substantial efforts to this task: the need for the libertarian paternalistic sleights of hand would be reduced; the effectiveness of the decisions of the rational self would be improved; last by not least, individuals would be treated with greater respect for their decisional autonomy. Yet libertarian paternalists hardly ever ask themselves how socially alterable the limits to rationality are, and seem to fall in the innate-bounds-to-rationality camp. As a result, suggestions as to how to improve the supposed cognitive shortcomings of individuals are conspicuous by their absence in their programme.

This is all the more surprising, because Sunstein himself (Sunstein, 2013), in his review of the work on scarcity by Mullainathan and Shafir, points out that 'feeling of scarcity [...] puts people in a cognitive tunnel, limiting what they are able to see. It depletes their self-control. It makes them more impulsive and a bit dumb. *What we often consider a part of people's basic character* – an inability to learn, a propensity to anger and impatience – *may well be a product of their feeling of scarcity.*'[83] (I discuss this aspect below.)

[81] see,eg, the discussion in Mitchell (2004)
[82] Nisbett, R, (2009)
[83] page 47, emphasis added.

8.3 Confidence in One's Thinking Abilities

The extent of the boundedness of people's rationality actually raises rather subtle questions, because one important result of Professor Nisbet's studies is the strong link between having confidence in one's intellectual abilities and reasoning well. Indeed, simply telling people that their intelligence is something malleable that they can shape – rather than a fixed an unchangeable endowment or an irreversible feature of their person, such as their height – is a crucial ingredient in bringing about an improvement in their thinking abilities. Most surprisingly, this turned out to be true even for mathematical skills – skills, that is, that are often regarded as intrinsically innate.

There is an important verso to this coin. If telling people that their logical and rational abilities can be improved helps them become better thinker, then one should question the extent to which emphasizing people's cognitive deficiencies – as libertarian paternalists never tire of doing in order to justify their measures – can become a self-fulfilling prophecy and can contribute to the problem they set out to fix. When a respected writer for *The Economist* opens an article about libertarian paternalism with the sentence "human beings are fallible: lazy, stupid, greedy and weak"[84], the average reader may well feel that, cognitively speaking, she may just as well give up. Looking at a number of different social contexts, it would not be the first time that this self-fulfilling prophesying has happened. This aspect is subtle and important, and deserves some further discussion.

8.4 Self-Fulfilling Prophecies

Libertarian paternalists often point to neurophysiological studies that suggest that, when reaching 'impulsive' and 'well-reasoned' (read, System-I or System-II) decisions, individuals appear to employ distinct areas of the brain. See, eg, the evidence reported in Sunstein (2005). Presumably, these findings are quoted by the libertarian paternalists to show how 'innate', 'hard-wired' and 'biologically rooted' their System-I/System-II distinction is. This 'evidence', however, must be treated with a lot of care. The idea that specific parts of the brain are immutably dedicated to certain functions has been put in question by a number of studies, such as those conducted by Nobel-prize winner Eric Kandel (2006).[85] The emphasis of these new studies has been on the *adaptability* of the brain. The careful observation of what happens in the brains of victims of stroke, for instance, has provided a very revealing insight into the way the brain works, and about its plasticity in particular. Important functions such as speech or motion have been observed to be recoverable by stroke victims, not because the damaged parts of the brain become 'reactivated', but because new neural pathways are forged, that carry out the lost functions. 'Plasticity' is the term used for this ability of the brain to 'rewire itself'. This ability has been

[84] See, eg, Bagehot in *The Economist*, 26 July, 2008, *Wink, Wink*

[85] see pages 198-202 and 204-207. See also Wexler (2006) and, for a non-technical discussion, Carr (2010).

observed to be neither limited to young age, nor to post-traumatic conditions. Quite simply, training and repetition, even relatively late in life, can effect a substantial redesign of which function are carried out by given parts of the brain.

Furthermore, the 'circuitry' of specific parts of the brain can be enhanced and literally grow with repeated use. For instance, in an often-quoted study (Maguire, Gadian, and Johnsrude (2000)) important changes were recorded in the brain of London cab drivers as they have switched from relying on their memory (their famous 'knowledge') to GPS navigation systems.[86]

Whether we tackle a certain task (such as making a decision) by making use of one part of the brain or another is therefore to some extent due to how we learn (or re-learn) to tackle a problem. Therefore, it is at least reasonable to speculate that if we become accustomed to 'reason out' the terms of a decisional problem, the associated (System-II) neural pathways are reinforced, and will be more readily used when we are faced with the next decisional task. If, on the other hand, we are not encouraged (by training or by the setting) to make well-reasoned choices, neural pathways in different part of the brain (loosely speaking, let's call then the pathways in the 'emotional parts of the brain') become reinforced, and will be more readily used next time. 'Neurons that fire together are wired together'.[87]

There is a flip side to this neuroplasticity. Once a set of neural pathways has been established, moving out of this too-well-ploughed furrow can be very difficult. Once a set of neural connections has been activated, by default the brain tries 'to keep it activated' and to reuse it to carry out the same tasks.[88] Unfortunately, there is no guarantee that the 'furrow' that has been established should be anything better than a very-local efficiency maximum. Furthermore, the process by means of which we learn to perform a given set of activities can be very path-dependent. As a result, we can get easily stuck in very sub-optimal ways of doing things. In particular, in bad ways of making decisions.

8.5 'Scarcity' and Its Effects on Decision-Making

An important observation: in his favourable review of *Scarcity* by Mullainathan and Shafirm, Sunstein (2013) asks the question of what can be done to mitigate the negative effects of scarcity (of money, time, etc) on the quality of decision-making: "Short of creating widespread abundance, can anything be done to reduce the harmful effects of scarcity?"[89]. With his gift for telling or retelling an engaging story , he mentions the repeated "wheels-up" crashes in WWII, 'which occurred when [bomber] pilots, upon landing, retracted the wheels rather than the flaps.'[90] After noticing how the problem was limited to pilots flying B-17s and B-25s, (for which the flap and wheel controls looked almost identical and

[86] See also Carr (2010), pages 32-33
[87] Carr (2010), Chapter 2.
[88] see Doidge, (2007).
[89] Sunstein (2013), page 48.
[90] ibid, page 48.

were placed side by side), the authors of *Scarcity* conclude (apparently seconded by Prof Sunstein) that '[p]ilot error turned out to be cockpit design error. A small change in the controls was enough to eliminate the problem.' Now, if this was the case, the example shows little about the detrimental effect of stress and scarcity on the quality of decision-making. But, probably, the lesson to be learnt is that poor design, coupled with stress, was giving rise to terrible outcomes.[91]

However, the 'fix' (simple, cheap and effective design improvement) is eagerly seized upon (it is not clear whether by Mullainathan and Shafirm, by Sunstein, or by both), and extrapolated as a possible solution to scarcity-induced cognitive biases: 'One possibility is to make certain outcomes automatic, so that people do not have to think about them at all. For example, many workers are busy, and they do not take the time to sign up for pension plans. In the United States, numerous employers have recently adopted automatic enrollment plans... [...] Another approach involves simple reminders. '[92]

If I read the review by Prof Sunstein correctly, these fixes are offered as a way to improve the decision-making of an individual by lightening her cognitive burden, not as a way to by-pass her decision-making process; they are, in this respect, enabling, not circumventing. Two objections can, however, be raised.

The first is that the interventions Sunstein proposes in his review include 'nudges' such as automatic enrollment in pension plans. But this is hardly a 'decisional distraction' (such as whether to discontinue the subscription to *Popular Mecahnics* we impulsively took out two years ago). Arguably, this is one of the decisions we *would* like to have decisional bandwidth to engage in. Once again, I feel that the libertarian paternalistic programme lacks a clear criterion telling us where it is 'fair game' to take away (or short-circuit) the full decision-making process from an individual, and where it is not.[93]

The second objection is that, as *Scarcity* cogently argues, not all bad decision come from having too much on one's plate – poverty is one major factor. But, if we really want to address this cause of poor decision-making, it is not necessary to create 'widespread abundance' (as Prof Sunstein writes) – a task which is presented as, and probably is, well-nigh impossible, and about which we should therefore not devote too much attention. Achieving widespread abundance may be impossible, but a substantial improvement of the most serious conditions of social and economic deprivation[94] *can* be fostered, for instance, by implementing more redistributive taxation regimes, or an increase in minimum wage; or, if worrying about how to cope with a possible illness makes us bad decision-makers,

[91] Prof Susntein, in a later passage recognizes as much: 'it might not be so illuminating to apply the term "scarcity" to a pilot struggling with a poorly engineered cockpit". (ibid, page 49). Indeed. However, by the point in the article where the qualification is introduced, the example, that lends itself perfectly to 'qucik, cheap and effective fixes' is alreday lodged in the memory of the reader.

[92] ibid, page 49

[93] This concern is also linked to my concerns about the benevolvence of the social architect, which I discuss in the following.

[94] The evidence presented in *Scarcity* shows a clear connection between cogntive impairement and *serious* deprivation. A 'bit of scarcity', the Authors contend, may actually be *good* for decision-making.

universal health insurance should be proposed. However, these are not quick, cheap and easy fixes. So, it seems to me that the suggestions advocated by a libertarian-paternalistic approach to the 'scarcity' problem either are welcome but minor palliatives ('sending text reminders'), which fail to go to the root of the solution to the poor decision-making process (teaching or *enabling* people to think well); or, if substantive in their scope, pose the same questions I have been trying to explore in this article.

8.6 Quick Fixes versus Difficult Choices

What are the implications of these findings for the libertarian paternalistic programme?

The first observation is that individuals can *learn (or be better enabled) to become* better thinkers. More subtly and more importantly, the same studies also suggest that *not* encouraging individuals to make conscious, deliberate, well-reasoned choices can reinforce the System-I neural wiring of the brain. And the more decisions are reached by, say, exploiting a lack of understanding of the effect of inflation (think of the Save-More-Tomorrow programme), or by relying on the inert acceptance of a default option, the less the neural pathways associated with our reflective mode of thinking are activated. The more, this means, our brain will *become* ill-equipped to reason rationally and critically about the next choice.

If our critical senses are not exercised and engaged, (and the attending neural pathways are re-routed away) what can we, as individuals and citizens, expect and hope for? For more and more nudges engineered by a benevolent libertarian paternalistic philosopher king to do on our behalf more and more of our deciding? This is not a prospect that I would welcome. (And it is, by the way, a version of liberalism that Mill would flatly refuse to recognize as such). It is in this sense that the cognitive deficiencies of individuals can truly become, by their systematic exploitation, a self-fulfilling prophecy.

Admittedly, training people to think better is unlikely to lend itself to quick-fix solutions. The libertarian paternalists could therefore be well justified in maintaining that their nudges are an *interim* measure until farther-reaching solutions can be given time to show their effectiveness. I have sympathy for this 'tactical' view, but this tactical interpretation were correct, it would greatly reduce the importance of libertarian paternalism.

9 Public Choice Analysis

Libertarian paternalists seem never to doubt that the intentions of the government (the choice-architect-in-chief, as Thaler and Sunstein (2008) say) is always benevolent, in the sense that it only has at heart only the welfare of the population.[95] This position neglects the insights provided by the schools that go

[95] Such a belief can be either naive, or rather sophisticated. If the latter, libertarian paternalists probably subscribe to the view that in a democratic society there are enough checks

under the name of public choice analysis, and of agency cost analysis.

Public choice analysis 'emerged as a reaction against the Pigovian, Keynesian, and standard political science approaches that (implicitly or explicitly) assumed that the primary concern of government officials is to make choices in the best interests of society as a whole – a benevolent public approach...' Somewhat surprisingly, this essential aspect of government interventions has received until relatively recently very little attention in the economic literature: while economic theory has been tackling over almost one hundred years more and more instances of 'market failure' and externalities, the 'imperfections' originating from the self-interested actions of the government have until relatively recently been barely examined[96]. Buchanan and Tullock, pioneers in this field, (1970) described their new approach to the study of the role of government as follows: '[i]n barest summary, we may say that 'government' is analyzed not as an organization that pursues some 'public interest' or other higher goals, but rather as a somewhat mundane machine for achieving the concrete desires of individual human beings. The central and pervasive research objective in Public Choice is that of understanding and explaining how the desires of citizens are transformed or translated into the observed outcomes of the political process.'

A second aspect of public choice-making is worth pointing out. Jensen and Meckling (1976) analyzed (mainly in the context of a firm, but their analysis has a much farther reach) the ability to align the actions of an agent with the desires of a principal. Their conclusions were that, in general, the agent's decisions and the decisions of the principal will be different*, even if monitoring and bonding by the principal and the agent, respectively, were optimal*: *even in the presence of optimal arrangements* between an agent and a principal (in our case, the choice engineer and the nudged individual), there remains an unavoidable wedge between the choices that the individual would make and the choices 'suggested' by the libertarian paternalist. In the analysis by Jensen, Meckling and others this discrepancy would persist even if all the actors were perfectly rational – more precisely, it would arise *exactly because* all the actors are perfectly rational utility maximizers[97].

In the light of this discussion we can now ask the following question: how can the libertarian paternalists justify their faith in the benevolence of the government? A benevolent politician should nudge voters towards the choices that the voters themselves (or, rather, their System-II) would make. When only one self is at play, the mechanisms whereby citizens ensure that politicians act in

on the actions of politicians to ensure that their policies are, often if not always, broadly in line with the best interest of the voters – if not, the argument goes, they would fail to be re-elected. From this perspective one does not have to rely on the personal benevolence of the individual politician, but simply on the effectiveness of the controls the electorate can place on the action of the government.

[96] "While economists refine the theoretical condition for perfect competition down to the last gnat's eyebrow, the gigantic problems of economic choice and allocation within our political institutions are left largely to political scientists and to a few economists working on defense-related problems", Downs (1964:88) in Medema (2009), page137.

[97] 'We retain the notion of maximizing behaviour on the part of all individuals in the analysis that follows' write Jensen and Meckling (1976).

their (the citizens') interest, are monitoring and voting. The process is fraught with difficulties, but at least there *may* be a mechanism to align (acceptably, if not perfectly) the interests of the electorate and the politicians. However, the alignment mechanism based on voting can only be expected to be effective if the System-II self drives the voting choice. But the libertarian paternalists doubt that this is always, or even often, the case. Indeed, Thaler and Sunstein report that "[i]t is possible to forecast the outcome of congressional elections with frightening accuracy simply by asking people to look quickly at pictures of the candidates and say which one looks more competent. These judgements, by students who did not know the candidate predicted the winner two thirds of the times! (Toderov et al (2005); Benjamin and Shapiro (2007))..."[98]

If indeed voting choices were so fickle and so fundamentally driven by the irrational System-I self, then there could be no hope that citizens could achieve optimal alignment via their monitoring and voting behaviour of the action of the politicians with their (the citizens') interests. In the absence of such alignment mechanisms, politicians would face no restraints from pursuing their ideological agenda, their pork-barrel policies and, even more brazenly, their naked personal interest. Not only would the faith in a benevolent choice-architect-in-chief be shattered, but one could almost certainly count on the fact that the action of politician would be *significantly* misaligned with the interests of the electorate. So, the faith in a benevolent nudger is put in question by the very libertarian paternalistic views of the two selves, and of what drives voting behaviour.

10 Conclusions

In this article I have articulated specific lines of criticism of libertarian paternalism. I would like to conclude with some more general considerations.

I have often raised doubts about the libertarian credentials of the libertarian paternalistic set of proposals. This does not mean, however, that I believe that the libertarian solution is always the most desirable one. On the contrary, I believe that in some instances a paternalistic approach can be appealing. For instance, I feel that a form of paternalism is justified is the overcoming of what I call information barriers. Overcoming information barriers is by no means the same as the passive provision of information. It is much more akin to fostering cognitive and intellectual growth. The barriers I refer to are the impediments that stand in the way of an individual's discovering and developing her tastes

[98] *Nudge*, page 20, footnote. I not in passing that Thaler and Sunstein do not elaborate on the conclusions that one should draw from these results., but if the libertarian paternalists truly believed these results to be robust and of general validity, one very natural conclusion would be the end *tout court* of the democratic programme. And there would be nothing 'libertarian' left to their programme: what is the value of free choice when it is so arbitrary, fickle and easy to manipulate? If these results really could truly to be extrapolated and interpreted as an indication of how the electorate make their choices, the programme a real believer in democracy should engage in with urgency is the overcoming, not the manipulation, of these cogntive weaknesses.

and capabilities. There are many activities that can deeply enrich the human experience. For instance, some of these profoundly moving activities involve the enjoyment of books, music, mathematics, theatrical performances, poetry etc. These experiences, at one's first encounter, may appear difficult, alien or even outright unpalatable. As a consequence, a freely choosing individual who has not been exposed at considerable length to these experiences cannot know if she will be moved by them – and, often, can hardly imagine that she will. One of the conditions for the libertarian approach to choosing to be defensible is the completeness of information. However, becoming 'informed' as to whether one will like opera, or Homer, or Kolmogorov probability theory is very different from becoming informed about the best car available for one's need. To some extent, the individual can only become 'informed' by 'doing'. When this 'doing' can be, at the beginning, disconcerting, difficult or even outright unpleasant, the information ends up not being gathered, and the ultimately 'better' choice ends up not being made.

I therefore believe that one important responsibility of a government is to help its citizens overcome these information barriers. Subsidies to education, theatres and opera, tax breaks for books, an unashamedly Reithian broadcasting policy and many other paternalistic measures are possible ways to remedy this state of affairs. When it comes to broadcasting, the alternative is the experiential waste land of the hundreds (thousands?) of freely chosen channels available on American TV. Given that American viewers on average spend hours per day watching TV, one can conclude that the viewers must 'like' what they see. And since, from a revealed-preferences perspective, they are certainly maximizing their utility function, from a libertarian point of view it is difficult to see what I am complaining about. The language of maximization, however, suggests precisely what the problem may be: the viewers may get stuck in a very shallow local maximum. They just do not know, *and are not given the means to discover*, that, if they were enabled to overcome a 'potential barrier', a much higher maximum may be available. With Keynes, I feel that the purpose of gaining economic and material comfort is for us to be able to enjoy a more rewarding life. If a long and well-provided-for retirement is spent watching *Big Brother*, I cannot help feeling that a big opportunity has been lost.

This is but an example. There are many other areas where I think, with the libertarian paternalists, that paternalistic interventions should be at the very least considered as a very strong alternative to libertarian *laissez faire*. It is well-known, of course, that serious problems attend unchecked or overenthusiastic paternalistic interventions.[99] The existence of these well-known problems does not necessarily imply, however, that inaction (Coase's 'fourth option': *do nothing*) is always the best alternative. As long as the electorate can monitor the actions of the politicians acceptably well; as long as their actions are transparent and auditable; as long as an adequately effective process to vote an overreaching government out of office exists; as long as all these conditions are reasonably met a bold and clearly visible (and the more clearly visible, the

[99] I articulate a few in *TL*, page 250 and *passim*

better) paternalistic intervention *can* in my view be justified.

As I have argued at length, it is exactly here that I feel libertarian paternalism is found most wanting: the original means of intervention it prefers owe much of their effectiveness to their 'flying below the radar screen' of rational scrutiny. My preference is therefore for a 'muscular', bold, *and therefore easily visible* form of paternalism, that, much as a Popperian hypothesis, leaves itself open wide up to ready rejection by the electorate.

There is another important aspect of the libertarian paternalistic programme that makes me uncomfortable. I cannot help but feeling that the cognitive deficiencies the behavioural economists never cease to point out, while real, may be artfully overemphasized in their literature. Indeed, Vernon Smith has described their programme as a 'candidly deliberate search'[100] for 'identifying the ways in which behaviour differs from the standard model'[101], a search 'in what may be the tail of a distributions'.[102] The adaptive value of the imperfect mode of the System-I reasoning is also downplayed by the libertarian paternalists, to a point that our very existence as a biological species becomes an evolutionary paradox. As we have seen, studies in cognitive psychology suggest that emphasizing the inability of an individual to perform certain cognitive tasks strongly affects (negatively, of course) her ability at the task. And this has been shown to be the case even in case of cognitive skills such as mathematical reasoning that are often thought to be deeply 'hard-wried'.

In this respect the reason for my discomfort is more than 'methodological'. What I feel deeply uncomfortable about are the implications of inevitability and helplessness of this supposed hard-wiredness and the attending vicious circle of 'decisional surrender'. Indeed, given the incessant drumming by the behavioural economists and the libertarian paternalists about how bad we are at making choices, it would not be surprising if theirs became a self-fulfilling prophecy. And when these conclusions about human nature are reached, liberalism becomes a rather vacuous concepts, and loses the enriching and capability-enhancing features it has possessed at the very least since the work of Mill. A world in which the cognitive biases of individuals were truly as pervasive as the libertarian paternalists claim; where *every* way of presenting (framing) information had as deep an influence on the resulting choice as claimed; in which *nominal* freedom of reversion choice could be offered in the knowledge and expectation that only a tiny minority would avail itself of it – such a world could not be described as 'libertarian' in any recognizable way.

Much more interesting and promising (and heart-warming), instead, are the current research programmes that try to establish how individuals can be helped in reaching better decisions. For instance, I have mentioned in Section 00 those studies aimed at discovering which settings encourage concentration and deliberate thinking and reduce the likelihood of making mistakes. These studies should be contrasted with the libertarian paternalistic near-determinism, an attitude that becomes at times intellectually suffocating.

[100] Smith (2008), page 22

[101] Mullainathan and Thaler (2001), page 2.

[102] Smith (2008), page 22.

In a way, the programme of the libertarian paternalists becomes hollow and self-defeating exactly if the cognitive deficiencies they point to are as widespread as they claim. Libertarian paternalists, in fact, claim that they are all in favour of freedom of choice[103]. But what would be the *value* of this freedom if it were indeed systematically the case that electors make their choices in the same unreflective and superficial way as described in the study by Toderov at al (2005) discussed in Chapter 10, and quoted by Thaler and Sunstein[104]? (The title of this paper – *Inferences of Competence from Faces Predict Election Outcomes* – speaks more eloquently about my concerns than I could possibly articulate). If a benevolent social architect truly believed that these results, and the results of endless related cognitive-bias-centred 'experiments', were as robust and transportable to real-life situations as the libertarian paternalists claim, surely what needs immediate attention is a programme to teach citizens to make informed, as opposed to wanton, choices.

This supposed prevalence of human cognitive deficiencies goes hand in hand with the libertarian paternalistic idea of the two selves. Their literature implies a much-more-than-metaphorical meaning to this construct, and even 'neurological' evidence is presented to buttress it. But what would happen to the libertarian paternalistic programme if, instead, we regarded the two-selves picture as nothing more than a colourful metaphor? What would happen, more precisely, if we denied to the two selves a homuncular status? The Coasian alternative of allowing the conflicts between the two selves to be resolved at level of the unitary individual (Coase's second option of internalization within a 'firm') then appears more natural, and far more appealing. And this conclusion would brings up back, via a rather tortuous route, to the old libertarian preference, that people should be allowed to make their own choices, occasionally mistaken as these choices may be.

Even if the two-selves picture of the individual were objectively correct,

I doubt that it can explain as much as the libertarian paternalists claim it does. When I look at the behaviours that the libertarian paternalists would try to modify via their nudges, only rarely am I confronted with situations where impulsive, from-the-gut choices are at the root of the problem. Inspired by Thaler-and-Sunstein mastery at personalized anecdotes (which has made thousands of readers familiar with Caroline, the cafeteria manager, and Janet, the future retiree), I have introduced in Section 00 my own real-life character, the man at the corner shop who bought two bottles of beer, a bag of salt-and-vinegar crisps, a packet of cigarettes and a candy bar. Allow me to call him Mike. Now, if that bundle was going to be Mike's 'tea' (as the time of day of the purchase suggested), why did he not buy – for less money – say, a pint of

[103] So, for instance: '...Our proposals are emphatically designed to retain freedom of choice...' Thaler and Sunstein (2008)

[104] Toderov, A, Mandisodza, A, N, Goren, A, Hall, C, C, (2005), *Inferences of Competence from Faces Predict Election Outcomes*, Science, 308, 1623-26. See the discussion in Chapter 10.

skimmed milk, two bananas, a loaf of whole-grain bread and an apple? Was it because he had 'truly' wanted to make these 'healthy' purchases all along, but could not resist the temptation of the items placed in prominent positions at the check-out counter? Was it because he had not been informed of the health consequences of smoking? Did he not know that a banana is more nutritious than a packet of salt-and-vinegar crisps? I doubt it.

And I also doubt that Mike compared in his mind, say, an apple against the crisps, the beer against the milk, etc, and chose the 'unhealthy' alternative because of some form of trade-off (faulty and biased as it might have been) between the alternatives.

So why *did* Mike buy those items? Of course, nobody can really know, but if I were to venture a guess, the most convincing account of Mike's purchases is that making these choices was part of his cultural, social and perhaps even 'class' identity. These are the foods and drinks that his family, his workmates and his friends consume. As such, they effectively define and circumscribe the range of his possible choices *and preferences*. And these choices and preferences, of course, are not limited to food and drink, but touch the TV programmes Mike watches, the newspapers he reads, the leisure activities he engages in, etc. In this picture of choice formation, the universe of effectively available alternatives is therefore restricted to those that are socially and culturally 'acceptable' and 'available'.

Now, we may agree that many individuals could lead healthier, more economically comfortable and more rewarding lives if they made different choices than the ones we observe them making. However, if my social and cultural explanation of the formation of many 'irrational' choices is not too wide of the mark, indiscriminately adopting as an explanation for these poor choices a construct based on 'irrational' impulsiveness and on a variety of cognitive biases is unhelpful at best, and harmful at worst.

It can be harmful because if we convince ourselves that poor choice are just a 'technical accident' in the thinking process – the mind, as it were, tripping over some easily circumventable obstacles – we are naturally drawn to conclude that all is needed for good decision-making is rectifying these mistakes. We can then convince ourselves that our interventions boil down to narrowly 'technological' tasks (exploit this cognitive bias here, harness the power of framing there, and so on) for which no great justification is required.

In reality, choice adaptation has social, not biological, roots. Capability enhancement is a social and cultural process, not a technological one. If we convince ourselves that the 'faulty' choices of normal, competent adults are like those of impulsive children and 'giddy teenagers', we fail to explore and understand the reasons why certain patterns of 'poor' and self-limiting choices are so common; we begin to feel instead that, as responsible 'parents', we are almost duty-bound to intervene; and we end up concluding that no complex justification is needed for our intervention. [105] So, yes, cognitive biases are real,

[105] I am not stretching my parental interpretation here, because the 'infantile' view of im-

but they are neither as pervasive and evolutionarily ineffective as the behavioural economists maintain, nor do they offer as compelling an explanation of 'poor' and self-limiting choices as the libertarian paternalists claim.

The reader can, of course, disagree with my thoughts on paternalism and on what is truly required to improve the quality, not of outcomes, but of decision-making – two topics that I have on purpose only sketched with a very broad brush. However, I hope that the reasons I have articulated as to why libertarian paternalism does not live up to its claims of being a benign, 'safe' and unobtrusive 'real-Third-Way' alternative both to paternalism and liberalism remain valid.

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perfect decision making is a recurrent one among the libertarian paternalists: of course, the *children's* cafeteria is one of their better known pièces de résistance, but it is not the only one: when they address the topic of the desirability of letting people learning from their own mistakes, for instance, Thaler and Sunstein (2008) mention *children* drowning in a swimming pool, despite the obvious inappropriateness of using an example about children to justify, of all things, *pater*nalism.

'...How much learning is good for people? We do not believe that *children* should learn the dangers of swimming pools by falling in and hoping for the best...' emphasis added.

References and Bibliography

Ackley, G, (1971), Discussion of a paper by James Tobin and Walter Dolde, in *Comsumer Spnding and Monetary Policy: The Linkages*, Boston: Federal Reseve Bank of Boston

Akerlof G A, (1970), *The Market for "Lemons": Quality Uncertainty and the Market Mechanism*, The Quarterly Journal of Economics, Vol. 84, No. 3. (Aug., 1970), pp. 488-500

Akerlof, G A, Shiller R J, (2009), *Animal Spirits - How Human Psyhology Drives the Economy, and Why It Matters for Global Capitalism*, Princeton University Press, Princeton, NJ

Alpert, M, Raiffa, H, (1982), *A Progress Report on the Training of Probability Assessors* in *Judgment under Uncertainty: Heuristics and Biases.* ed. D. Kahneman, P. Slovic, A. Tversky, Cambridge University Press, New York

Al-Najjar, N I, Whitman, J, (2009), *The Ambiguity Aversion*, Economics and Philosophy, 25, 249-284

Anderson, J R, (1991), *The Adaptive Nature of Human Categorization*, Pshycological Review, 98, 409-429

Andreoni, J, Miller J H, (2002), *Giving According to GARP: An Experimental Test of Consistency of Preferences for Altruism*, Econometrica, 70, 2, 737-753

Archard, D, (1993), *Self-Justifying Paternalism*, Journal of Value Inquiry, 27, 341-352

Ariely, D, (2009), *Predictably Irrational - The Hidden Forces That Shape Our Decisions*, Second Edition, Harpers Collins Publishers

Arrow, K J, (1963), *Social Choice and Individual Values,* Yale University Press, Yale, New Haven, CT

Arrow, K J, (1969), *The Organization of Economic Activity: Issues Pertinent to the Choice of Market versus Non-market Allocation*, paper delivered at the Joint Economic Committe of Congress, availbale at

http://.msuweb.montclair.edu/~lebelp/PSC643IntPolEcon/ArrowNonMktActivity1969.pdf, last accessed 20 September, 2011

Axelrod, *The Evolution of Cooperation*,

Baker, M, Stein J C, Wurgler J, (2006), *The Effect of Dividends on Consumption*, National Bureau of Economic Reserach, Working Paper 12288, June 2006

Barberis, N, Santos T, Huang M, (2001), *Prospect Theory and Asset Prices*, Quarterly Journal of Economics, 116, 1-53

Bardsley, N, Cubitt, R, Loomes, G, Moffatt, P, Starmer C, Sugden, R, (2009), *Experimental Economics: Rethinking the Rules*, Princeton Unviersity Press, Princeton, NJ

Becker, G A, Posner, R A, (2009), *Uncommon Sense: Economic Insight from Marriage to Terrorism*, University of Chicago Press, Chicago, IL

Benartzi, S, Thaler, R H, *Heuristics and Biases in Retirement Savings Behaviour*, Journal of Economic Perspectves

Benartzi, S, Thaler, R H, (2002), *How much Is Investor Autonomy Worth?*, Journal of Finance, LVII, No 4, August, 1593-1616

Benjamin D, Sahpiro J, (2007), *Thin-Slice Forecast of Gubernatorial Elections*, Working paper, University of Chicago

Berg J, Dickhaut J, McCabe K, (1995), *Trust, Reciprocity and Social History*, Games and Economic Behaviour, 10, 122-142

Berg N, Gigerenzer, G, (2007), *Psychology Implies Paternalism? Bounded Rationality May Reduce the Rationale to Regulate Risk-Taking*, Social Choice and Welfare, 28, 2, 337-359

Binmore, K, (2007), *Does Game Theory Work?*, The MIT Press, Cambridge, Ma

Binmore, K, (2009), *Rational Decisions*, Princeton University Press, Princeton, New Jersey

Blackwell, L, Trzesniewski K, Dweck, C S, (2007), *Implicit Theories of Intelligence Predict Achievement Across an Adolescent Transition: A Longitudinal Study and an Intervention*, Child Development, 78, 246-263

Bossaert, P, Odegaard, B A, (2010), *Lectures on Corporate Finance*, Second Edition, World Scientific, Singapore and London

Bowles, S, Gintis, H, (2006), *The Evoltionary Basis of Collective Action*, Chapter 53 in *The Oxford Handbook of Political Economy*, Weingast and Wittman editors, Oxford University Press, Oxford

Brams, S J, (2008), *Mathematics and Democracy: Designing Better Voting and Fair-Decision Procedures*, Princeton University Press, Princeton, NJ

Brighton, H, Gigerenzer G, (2008), *Bayesian Brains and Cognitive Mechanisms: Harmony or Dissonance?*, Chapter 9, in Chater, M, Oaksford, M, (2008), Eds, *The Probabilistic Mind – Prospects for Bayesian Cogntive Science*, Oxford University Press, Oxford

Brown, C, and Krishna, A, (2004), *The Skeptical Shopper: A Metacognitive Account for the Effects of Default Options on Choice*, Journal of Consumer Research, 31, 529-539

Buchanan, J M, Tullock, G, (1970), Centre for Study of Public Choice National Science Foundation proposal, Buchanan House Archives.

Buller, D J, (2005), *Adapting Minds: Evolutionary Psychology and the Persistent Quest for Human Nature*, MIT Press, Cambridge, MA

Camerer, C, et al., (2003), *Regulation for Conservatives: Behavioral Economics and the Case for "Asymmetric Paternalism",* University of Pennsylvania Law Review, 151, 1211-1233

Campbell Mcconnel, R, Brue, L, Flynn, S M, (2008), *Microeconomics*, 18th edition, MacGraw Hill

Caplan, B, (2007), *The Myth of the Rational Voter*, Princeton University Press, Princeton, New Jersey

Carlin, B I, Gervais, S, Manso, G, (2010), *Libertarian Paternalism, Information Strategies and Financal Decision Making*, available at SSRN:http://ssrn.com/abstract=1570158, last accessed 18th Febraury, 2012.

Carr, N, (2010), *The Shallows – What the Internet Is Doing To Our Brains*, Norton, New York, NY

Chamley, C P, (2004), *Rational Herds – Economic Modelling of Social Learning*, Cambridge University Press, Cambridge, UK

Chater M, Oaksford M, (2008), Eds, *The Probabilistic Mind – Prospects for Bayesian Cogntive Science*, Oxford University Press, Oxford

Chief Medical Officer (2001), *The Removal, Retention and Use of Human Organs and Tissue from Post-Mortem Examination*, TSO

Choi, J J, Laibson, D, Madrian, B C, Metrick, A (2002), *Defined Contribution Pensions: Plan Rules, Participant Choices, and the Path of Least Resistance*, in James Proterba, editor, *Tax, Policy & the Economy*, 67-114

Churchland, P, (2011), *Braintrust – What Neuroscience Tells Us about Morality*, Princeton University Press, Princeton, New Jersey

Cochrane, J H, (2001), *Asset Pricing*, Princeton University Press, Princeton, NJ

Cochrane, J H, (2010), *Discount Rates*, University of Chicago and NBER working paper, http://faculty.chicagobooth.edu/ john.cocrane/research/Papers

Coase, R H, ([1976]) 1994, *Adam Smith's View of Man.*, in *Essays on Economics and Economists,* Chicago: University of Chicago Press, 95-116, quoted in Klein (2004)

Coase, R H, (1960), *The Problem of Social Cost*, Journal of Law and Economics, 3, October, 1-44

Crabtree, J, (2010), *Nudge Nudge, Wonk Wonk*, Financial Times, Notebook, page 10, August 10, 2010.

Crisp, R, (1998, reprinted 2004), Editor's introduction to Mill, J S, *Utilitarianism*, Oxford University Press

Czerlinsky, J, Gigerenzer, G, Goldstein, D G, (1999), *How Good Are Simple Heuristics?*, in Gigerenzer, G, Todd, P M, *Simple Heuristics That Make Us Smart*, 97-118, Oxford University Press, Oxford and New York and

http://www.scribd.com/doc/62298863/Czerlinski-How-Good-Are-Simple-Heuristics-Chapter-1999 last accessed 1st November, 2011.

Damasio, A, (1994), *Descartes' Error: Emotion, Reason and the Human Brian*, Avon Boks, New York

Dasgupta, P, (2008), *Discounting Climate Change*, Journal of Risk and Uncertainty, 37, 141-169

Dasgupta, P, Maskin, E, (2005), *Uncertainty and Hyperbolic Discounting*, The American Economic Review, Vol 95, no 4, 1290-1299

Demirer, R, Mau, R R, Shenoy, C, (2003), *Bayesian Networks: A Tool to Improve Portfolio Risk Analysis*, working paper, University of Kansas Business School

Dennet, D C, (2013), *Intuition Pumps and Other Tools for Thinking*, Allen Lane (Penguin Group), London

Dixit, A, K, and Pindyck, R S, (1994), *Investment Under Uncertainty*, Princeton Univeristy Press, Princeton, NJ

Doidge, N, *The Brain That Changes Itself*, Penguin Books, New York

Donner, W, Fumerton, R, (2009), *Mill*, Blackwell, Oxford

Downs, A, (1964), Review of *The Calculus of Consent*, Journal of Political Economy, 72, February, 87-88

Druckman J N, (2001), *The Implicatons of Framing Effects for Citizen Competence*, 23, Political Behavior, 225, 237

Dworking, G, (2006), *Paternalism*, in Edward N Zalta, Editor, *The Stanford Encyclopedia of Philosophy*

Edgewotrh, F Y, (1881), *Mathematical Psychics: An Essay on the Application of Matehmatics to the Moral Sciences*, Kegan Paul, London

Edwards A G K, Elwyn G J, Covey J, Mathews E, Pill R, (2001), *Presenting Risk Information: A Review of the Effects of "Framing" and other Manipulations on Patient Outcomes*, Journal of Health Communications, 6, 61-82

Elliott, A, (2004), *Fair Banking: The Road to Redemption for UK Banks*, CSFI (Centre for the Study of Financial Institutions), London

Ellsberg, D, (1961), *Risk, Ambiguity, and the Savage Axioms*, The Quarterly Journal of Economics, 75, (4), 643-669

Epstein L G, and Zin S, (1989), *Substitution, Risk Aversion and the Temporal Behaviour of Consumption and Asset Returns: A Theoretical Framework*, Econometrics, 57, 937-969

Etzioni, A, (2008), *Disclosure Is Not Enough*, The Huffington Post, Posted Septermber 18th, 2008, 04:31 PM (EST), last accessed 17th February, 2009

Feller, W, (1950), *An Introduction to Probability and Its Applications*, 3rd edition, Vol 1, Wiley India

Fodor, J, (2011), *Fire the Press Secretary*, London Review of Books, 28 April 2011, 24-25.

Forgas, J P, Williams K D, and Von Hippel, W, (eds) (2003), *Social Judgements*, Cambridge University Press, Cambridge

Frank, R H, (2004), *What Price the Moral High Ground*, Princeton University Press, Princeton, New Jersey

Frazzetto, G, (2013), *How We Feel: What NeuroScience Can – and Can't – Tell Us About Our Emotions*, Doubleday, London

Friedman, D, (2008), *Morals and Sentiments – An Evolutionary Account of the Modern World*, Palgrave MacMillan, New York, New York

Friedman, M, (1953), *The Methodology of Positve Economics*, in Essays in Positive Economics, University of Chicago Press, Chicago

Friedman, M, (1962 – 1982), *Capitalism and Freedom*, Univeristy of Chicago Press, Chicago and London

Frydman R, Goldberg M D, (2007), *Imperfect Knowledge Economics: Exchange Rates and Risk*, Princeton University Press, Princeton, NJ

Froyen, R T, (1996), *Macroeconomics – Theories and Policies*, fifth edition, Prentice Hall, Upper Saddle River, New Jersey

Geanakoplos, J, (1996, 2005), *Three Brief Proofs of Arrow's Impossibility Theorem*, Cowles Foundation Discussion Papers, no 1123R3 - 1996, available at http://cowles.econ.yale.edu/P/cd/d11a/d1123-r3.pdf, last accessed 10 September 2011, published in *Economic Theory* (2005), 26, 1, 211-215

Gigerenzer, G, (2002), *Reckoning with Risk: Learning to Live with Uncertainty.* London, Penguin, 2002 (Published in the United States as *Calculated Risks: How to Know when Numbers Deceive You.* New York: Simon & Schuster)

Gigerenzer, G, Brighton, H, (2009), *Homo Heuristicus: Why Biased Minds Make Better Inferences*, Topics in Cognitive Science, 1, 107-143

Gigerenzer, G, Edwards, A, (2003), *Simple Tools for Understandign Risk: From Innumeracy to Insight*, British Medical Journal, Vol 327, 27 September 2003, 741-744

Gigergrenzer, G, Hoffrage, U, (1995), *How to Improve Bayesian Reasoning Without Instructions. Frequency Formats*, Psychological Review, 102, 684-704

Gigergrenzer, G, Selten, R, (2002), *Rethinking Rationality*, in *Bounded Rationality – The Adaptive Toolbox*, Gigerenzer G and Selten R, Editors, MIT Press, Cambridge, Mass

Gigerenzer, G, and Selten R, (Editors), (2002), *Bounded Rationality – The Adaptive Toolbox*, MIT Press, Cambridge, Mass

Gilboa, I, (2010), *Rational Choice*, MIT Press, Cambridge, Mass and technical appendices available at http://mitpress.edu/rationalchoice

Gintis, H, (2009), *The Bounds of Reason – Game Theory and the Unification of the Behavioural Sciences*, Princeton University Press, Princeton, New Jersey

Gintis, H, Khurana, R, (2008), *Corporate Honesty*, in *Moral Markets*, Zak, P J, editor, Princeton University Press, Princeton

Glaeser, E L, (2006), *Paternalism and Psychology*, University of Chicago Law Review, January, 73: 133-156

Goodin, R, (1993), *Democracy, Preferences and Paternalism*, Policy Sciences, 26, 22-247

Gray, J, (2000), *Two Faces of Liberalism*, The New Press, New York, NY

Gray, J, (1996, 2013), *Isaiah Berlin – An Interpretation of His Thought*, Princeton University Press, Princenton, NJ

Green, G, (2009), *A Nudge in the Right Direction*, Metro, 6 May, 2009, page 16

Griffiths, T L and Tenenbaum J B, (2005), *Optimal Predictions in Everyday Cognition*, Psychological Science 17(9):767–73

Gruber, J, Koszegi, B, (2001), *Is Addiction 'Rational'? Theory and Evidence*, Quartterly Journal of Economics, 116, 1261-1294

Grunwald, M, (2009), *How Obama Is Using the Science of Change*, Time, 13 April 2009, 28-32

Gutmann, A, (1987), *Democratic Education*, Princeton University Press, Princeton, New Jersey

HM Treasury, (2006), *Stern Review on the Economics of Climate Change*

Hagreaves Heapes, S P, Varoufakis, Y, (2004), *Game Theory – A Critical Text*, Second edition, Routledge, London

Hald, A, (1990), *A History of Probability and Statistics and Their Applications Before 1750*, John Wiley and Sons, New York

Hardin R, (2009), *How Do You Know? The Economics of Ordinary Knowledge*, Princeton University Press, Princeton, NJ

Hauser, M D, (2006), *Moral Minds – How Nature Designed Our Universal Sense of Right and Wrong*, Harper Collins, New York

Hayek, F A, (1937), *Economics and Knowledge*, Economica, 4, 13, 33-54

Hayek, F A, (1940), *The Competitive Solution*, Economica, 7, 26, 125-149

Hayek, F A, (1944), *The Road to Serfdom*, University of Chicago Press, Chicago, IL

Henrich, J R, Boyd, R, Bowles, S, Camerer, C, Fehr, E, Gintis, H, (2004), *Foundations of Human Sociality: Economic Experiments and Ethnographic Evidence from Fifteen Small-Scale Societies*, Oxford University Press, New York, NY

Hens, T, Rieger, M O, (2010), *Financial Economics: A Concise Introdution to Classical and Behavioral Finance*, Springer Verlag, Heidelberg, Berlin

Herring, J, (2008), *Medical Law and Ethics*, Oxford University Press, Oxford

Holt C A, (1995), *Industrial Organization: A Survey of Laboratory Research*, Princeton University Press, Princeton, NJ

Holt C A, Langan L, Villamil A, (1986), *Market Power in an Oral Double Auction*, Economic Inquiry, 24, 107-123

Human Tissue Act (2004), UK Department of Health

Ippolito, R A, (2005), *Economics for Lawyers*, Princeton University Press, Princeton, New Jersey

Jacob, E, (2011), *Recipe for a Healthier Office*, Financial Times, November, 3, 2011, page 14

Jensen, M C, (2003), *A Theory of the Firm – Governance, Residual Claims and Organizational Forms*, second edition, Harvard University Press, Harvard, Mass

Jensen, M C, Meckling W H, (1976), *Theory of the Firm: Managerial Behaviour, Agency Cost and Ownership Structure*, Journal of Financial Economics, 3, no 4, 305-350

Jensen, M C, Meckling W H, (1994), *The Nature of Man*, Journal of Applied Corporate Finance, 7, 2, (summer), 4-19

Jensen, M C, Smith C W (jr), (1985), *Stockholders, Managers and Creditor Interests: Applicatin of Agency Theory*, in *Recent Advances in Corporate Finance*, Altman, E I, Subramanyan M G, editors, Irwin, Homewood, Ill, 93-131.

Johansson, P-O, (1991), *An Introduction to Modern Welfare Economics*, Cambridge University Press, Cambridge, UK

Jolls, C, Sunstein, C, (2005), *Debiasing Through Law*, University of Chicago, John M Olin School of Law and Economics Working Paper no 225, second series

Johnson, E J, Golsdtein, D, (2003), *Do Defaults Save Lives?*, Science, Vol 302, 21 November 2003, 1338-1339

Kahneman, D, (2002), *Maps of Bounded Rationality: Psychology of Behavioural Economics*, lecture delivered in Stockholm, Sweden, on December 8, 2002, upon reception of the Bank of Sweden Prize in Economic Sciences in Memory of Alfred Nobel

Kahneman, D, Tversky, A, (1979), *Prospect Theory: An Analysis of Decisions Under Risk*, Econometrica, 47, 263-291

Kahneman, D, Slovic, P, Tversky, A, (1982), (Editors), *Judgement under Uncertainty: Heuristics and Biases*, Cambridge University Press, Cambridge, UK

Kahnneman, D, Tversky, A, (1972a), *On Prediction and Judgement*, ORI Research Monograph, 12 (4)

Kahnneman, D, Tversky, A, (1972b), *Subjective Probability: A Judgement of Representativeness*, Cognitive Psychology, 3, 430-454

Kahnneman, D, Tversky, A, (1973), *On the Pshycology of Prediction*, Psychological Review, 80, 237-251

Kahnneman, D, Tversky, A, (1979a), *Intuitive Prediction: Biases and Corrective Procedures*, TIMS Studies in Management Science, 12, 313-327

Kahnneman, D, Tversky, A, (1979b), *Prospect Theory: An Analysis of Decision Under Risk*, Econometrica, 47, 263-291

Kahneman D, Tversky A, (2000), (Editors), *Choices, Values, and Frames*, Cambridge University Press, Cambridge, UK

Kendal, E R, (2006), *In Search of Memory: The Emergence of a New Science*, Norton, New York

Keynes, J K, (1926), *The End of Laissez-Fair*, Hogarth Press, London

Kimbrough, E O, Smith, V L, Wilson B J, (2008), *Building the Market*, in *Moral Markets*, Zak, P J, editor, Princeton University Press

Klein, D B, (2004), *Status Quo Bias* – A Comment on R H. Thaler and C. R. Sunstein, 2003, *Libertarian Paternalism*, American Economic Review (AEA Papers and Proccedings) 93, (2), 175-79, Economics Journal Watch, Vol 1 , Number 2, 260-271

Knill D C, Pouget A, (2004), *The Bayesian Brain: The Role of Uncertainty in Neural Coding and Computation*, Trends in Neuroscience, 27, (12), 712-719

Korobkin, R, (2009), *Libertarian Welfarism*, California Law Review, Vol. 97, No. 6, 2009 and UCLA School of Law, Law-Econ Research Paper No. 09-09

Kreps, D M, (1988), *Notes on the Theory of Choice*, Westivew Press, Boulder, Co

Kreps, D M, (1990), *A Course In Microeconomic Theory*, Harvertser Wheatsheaf Publishing, New York

Kurzaban, R, (2010), *Why Everyone (Else) Is a Hypocrite – Evolution and the Modular Mind*, Princeton University Press, Princeton, New Jersey

Kymblica, W, (2002), *Contemporary Political Philosophy*, Second Edition, Oxford University Press, Oxford

Laibson, D, (1997), *Golden Eggs and Hyperbolic Discounting*, Quantitative Journal of Economics, 62, 443-477

Lange, O, (1936), *On the Economic Theory of Socialism*, Review of Economic Studies, 4, 1, 53-71

Layard, R, (2005), *Happiness: Lessons from a New Science*, Penguin, London

LeDoux, J, (1996), *The Emotional Brian*, Simon and Shucster, New York

Leherer, J, (2010), *Does Moral Action Depend on Reasoning?*, debate in www.templeton.org/reason, last accessed 20th June, 2010.

List J A, Millimet, (2008), *The Market: Filter of Irrationality and Catalyst of Rationality*, The B. E. Journal of Economic Analysis and Policy, Vol 8, Issue 1 (Frontiers), Article 47, Available at: http://www.bepress.com/bejeap/vol8/iss1/art47

Madrian, B, and Shea, D, (2001), *The Power of Suggestion: Inertia in 401(k) Participation and Savings Behavior*, Quarterly Journal of Economics, 116, 1149-1187

Marglin, S A, (2008), *The Dismal Science – How Thinking Like an Economist Undermines Community*, Harvard University Press, Cambridge Massachussets and London, England

Maclean, A, (2001), *A Crossing of the Rubicon on the Human Rights Ferry*, Medical Law Review, 64, 775

Maguire, E A, Gadain, D G, Johnsrude, I S, (2000), *Navigation-Related Structural Changes in the Hippocampi of Taxi Drivers*, Proceedings of the National Academy of Sciences, 97, 8, April, 4398-4403

Martignon, L, (2002), *Comparing Fast and Frugal Heuristics and Optimal Models*, in *Bounded Rationality – The Adaptive Toolbox*, Gigerenzer G and Selten R, Editors, MIT Press, Cambridge, Mass

McGinn, C, (2011), *Can the Brain Explain Your Mind?*, New York Review of Books, March 24, 2011, 32-35

McKenzie, (2006), *An Enigne, not a Camera*, MIT Press, Cambridge, Mass, USA

McKenzie, C, Liersch, M, and Finkelstein, S, (2006), *Recommendations Implicit in Policy Defaults*, Psychological Science, 17, 414-420

Medema, S G, (2009), *The Hesitatnt Hand – Taming Self-Interest in the History of Economic Ideas*, Princeton Univerity Press, Princeton, New Jersey.

Mill, J S, (1859), *On Liberty*, Ticknor and Fields, Boston, Ma

Mill, J S, (1871), *Principles of Political Economy*, 7th edition, Longmans, Green, London

Mitchell, G, (2002), *Taking Behavioralism Too Seriously? The Unwarranted Pessimism of the New Behavioral Analysis of Law*, 43, WM.& Maryland Law Review, 1907, 1961-63, 2005-11

Mitchell, G, (2004), *Libertarian Paternalism Is an Oxymoron*, FSU College of Law, Public Law Research Paper No. 136; FSU College of Law, Law and Economics Paper No. 05-02

Mitchell, O, Mottola, G, Utkus, S, and Yamaguchi, T, (2009), *Default, Framing, and Spillover Effects: The Case of Lifecycle Funds in 401(k) Funds*, National Bureau of Economic Research, Working Paper.

Mullainathan, S, Thaler R, (2001), *Behavioural Economics*, in Smelser N J and Baltes, P B, editors, *International Encyclopedia of the Social and Behavioural Sciences*, Elsevire, Oxford

Mullainathan, S, Shafir, E, (2013), *Scarcity: Why Having So Little Means So Much*, Times Books

New, B, Le Grand, J, *Paternalism*, Princeton Univeristy Press, Woostock, UK, and Princeton, NJ, USA, forthcoming

Nisbett, R, (2009), *Intelligence and How to Get It – Why Schools and Cultures Count,* W.W.Norton, New York and London

Nordhaus W, (2007), *The Stern Review on the Economics of Climate Change*, Journal of Economic Literature, 45, (September), 686-702

Nordhaus W, (2008), *A Question of Balance: Weighing the Options on Global Warming Policies*, Yale University Press, New Haven and London

Nozick, R, (1974 - 2009), *Anarchy, State and Utopia*, Blackwell Publishing, Malden, MA, and Oxford, UK

Nussbaum, M C, (2000), *Women and Human Development*, Cambridge University Press, Cambridge, UK

O'Donoghue T, Rabin, M, (1999a), *Doing It Now or Later*, American Economic Review, 89, 1, March, 103-124

O'Donoghue T, Rabin, M, (1999b), *Incentives for Procastinators*, Quarterly Journal of Economics, 114, 3, 769-816

O'Donoghue T, Rabin, M, (2000), *The Economics of Immediate Gratification*, Journal of Behavioural Decision-Making, 13, 2, 233-250

O'Donoghue T, Rabin, M, (2001), *Choice and Procastination*, Quarterly Journal of Economics, 116, 1, 121-160

O'Donoghue T, Rabin, M, (2003), *Optimal Sin Taxes*, unpublished manuscript, University of California at Berkeley, 3, p. 2.

Orsekes N, Conway E, (2010), *Merchants of Doubt: How a Handful of Scientists Obscured the Truth on Issues from Tobacco Smoke to Gloal Warming*, Bloomsbury, London

Paul, R L, Goodman, H, Merzenich, M, (1972), *Alterations in Mechanoreceptor Input to Braodman's Areas 1 and 3 ... after Nerve Section and Regeneartion*, Brain Research, 39, 1, April, 1-19

Pascual-Leone, A, Amedi, A, Fregni, F, Merabet, L B, (2005), *The Plastic Human Brain Cortex*, Annual Review of Neuroscience, 28, 377-401

Persson, I, (2008), *The Retreat of Reason – A Dilemma in the Philosophy of Life*, Oxford University Press, Oxford, UK

Petit-Zeman, S, (2010), *Giving People What They Want*, Prospect, July 2010, page 58

Pope, T M, (2004), *Counting the Dragon's Teeth and Claws: The Definition of Hard Paternalism*, Georgia State University Law Review, 20, 659-722.

Posner, R A, (2010), *The Crisis of Capitalist Democracy*, Harvard University Press, Cambridge, MA

Qizilbash, M, (2009), *Well-Being, Preference Formation and the Dangers of Paternalism*, paper on Economics and Evolution, Max Plank Institute, University of Jena, paper # 0918

Qizilbash, M, (2011), *Sudgen's Critique of Sen's Capability Approach and the Dangers of Libertarian Paternalism*, International review of Economics, 58, 21-42

Rabin, M (2000), Kahneman, D, and Tversky, A, (eds) (2000), *Choices, Values and Frames*, Chapter 11, Cambridge University Press, Cambridge, UK

Radner, R, (1995), *Economic Survival*, Nancy L Schwartz Lecture delivered May 1995 at the J L Kellog School of Management, Northwestern University.

Ralfe, J, (2009), *Pension Report Was Flawed*, Financial Times, page 11, FTfm section

Retained Organ Commission, (2004), *Response to Department of Health Consultation Report – Human Bodies, Human Choices*, Department of Health

Ridley, M, (2008), endorsement of Zak (2008)

Rothbard, Murray N, (1956), *Toward a Reconstruction of Utility and Welfare Economics*, in *On Freedom and Free Enterprise: Essays in Honor of Ludwig von Mises*, ed. M. Sennholz. Princeton, NJ: D. Van Nostrand, 224-262.

Rothbard, Murray N, (1962), *Man, Economy, and State*, Princeton, NJ: D. Van Nostrand.

Saint-Pual, G, (2011), *The Tyranny of Utility – Behavioral Science and the Rise of Paternalism*, Princeton University Press, Princeton, New Jersey

Samuelson, P A, (1937), *A Note on the Measurement of Utility*, The Review of Economic Studies, 4, 155-161

Samuelson, P A, (1938), *A Note on the Pure Theory of Consumers' Behaviour*, Economica, 5, and *A Note on the Pure Theory of Consumers' Behaviour: An Addendum, ibid*

Sauermann Selten,

Schelling, T C, (2006), *Strategies of Commitment*, Harvard University Press, Cambridge, Massachusetts

Schelling, T C, (1978), *Micromotives and Macrobehaviour*, Norton, New York and London

Schelling, T C, (1994), *Intergenerational Discounting*, Energy Policy, VoL 23, No. 4/5, 395-401

Schwartz, B, (2004), *The Paradox of Choice*, Harper Collins, New York, NY

Scoot, J C, (2008), *Seeing Like a State: How Certain Schemes to Improve Human Condition Have Failed*, Yale University Press, New Haven, Connecticut

Selten, R, (1991), *Evolution, Learning and Economic Behaviour*, Games and Economic Behaviour, 3, 3-24

Selten, R, (1998), *Features of Experimentally Observed Bounded Rationality*, European Economic Review, 42, 413-436

Sen, A K, (1973), *Behaviour and the Concept of Preference*, Economica, 40, August, 241-259

Sen, A K, (1977), *Rational Fools: A Critique of the Behavioural Foundations of Economic Analysis*, Philosophy and Public Affairs, 6, Summer, 317-344

Sen, A K, (1979), *Personal Utilities and Public Judgement: or What's Wrong with Welfare Economics?*, Economic Journal, September, 537-558

Sen, A K, (1980), *Equality of What?*, Tanner Lectures in Human Values, in McMurrin, S, editor, Cambridge University Press, Cambridge, UK

Sen, A K , (1982), *Choice, Welfare and Measurement*, Harvard University Press, Harvard, Mass

Sen, A K, (2008), *The Idea of Justice*, Allen Lane, Penguin Group, London

Sen, A K, (1999a), *Development as Freedom*, Anchor Books, Random House, New York

Sen, A K, (1999), *Rights Consequentialism*, in *Utilitarianism and Its Critics*, Glover, J, Editor, Macmillan, New York

Shiller R J, (2008), *The Subprime Solution*, Princeton University Press, Princeton, NJ

Shleifer A, (2000), *Inefficient Markets – An Introduction to Behavioural Finance*, Clarendon Lectures in Economics, Oxford University Press, Oxford

Shefrin H, Thaler R, (1988), *The Behavioural Life-Cycle Hypothesis*, Economic Inquiry, 24, 609-643

Sidgwick , H, (1897), *The Elements of Politics*, 2nd Edition, Macmillan, London

Sigmund, K, (2010), *The Calculus of Selfishness*, Princeton University Press, Princeton, NJ

Simon, H, A, (1956), *Rational Choice and the Structure of Environments*, Psychological Review, 63, 129-138

Simon, H, A, (1957), *Models of Man*, John Wiley, New York

Sinn, H-W, (2002), *Weber's Law and the Biological Evolution of Risk Preferences: The Selective Dominance of the Logarithmic Utility Function*, CESIFO working paper 770, September 2002, 1-28

Skidleski, R, Wigstroem, C W, (2010), *The Economic Crisis and the State of the Economy*, Palgrave MacMillan, New York

Skinner, Q, (2008), *Hobbes and Republican Liberty*, Cambridge University Press, Cambridge, UK

Skypala, P, (2010), *Let's Not Go Full Circle on Savings*, Financial Times, Monday, June 28th 2010, page 6, section FTfm

Smith, A, (1790, 2010), *The Theory of Moral Sentiments*, Penguin Classics, London

Smith, V, L, (1962), *An Experimental Study of Competitive Market Behaviour*, Journal of Political Economy, 70, 111-137

Smith, V, L, (1962), *Microeconomic Systems as an Experimental Science,* American Economic Review, 72, (December), 923-955

Smith, V,L, (2003), *Constructivist and Ecological Rationality in Economics,* American Economic Review, 93, 3, 465-508

Smith, V, L, (2008), *Rationality in Economics – Constructivist and Ecological Forms*, Cambridge University Press, Cambridge, UK

Stigler, G, J, (1961), *The Economics of Information*, Journal of Political Economy, 69, 191-214

Sugden, R, (2006), *What We Desire, What We Have Reason To Desire, Whatever We Desire: Mill and Sen on the Value of Opportunity*, Utilitas, 18, 1, 33-51

Sugden, R, (2008), *Why Inchoerent Preferences Do Not Justfy Paternalism*, Constitutional Political Economics, 19, 226-248

Sugden, R, (2009), *On Nudging: A Review of 'Nudge'*, International Journal of Economics and Business, 16, 3, 365-373

Sunstein, C, (1997), *The Behavioral Analysis of Law*, University of Chicago Law Review, Vol 64. No 4, 1175-1195

Sunstein C, (2005), *Laws of Fear – Beyond the Precautionary Principle*, Cambridge University Press, Cambridge

Sunstein, C, (2013), *It Captures Your Mind*, book review of *Scarcity: Why Having Too Little Menas So Much*, by Mullainatha and Shafir, New York Review of Books, September 26th-October 9th, 2013, Vol LX, No 14, pages 47-49

Sunstein, C, (2013b), *Government Shutdown Has a Simple Explanation*, Bloomberg Views, Oct 1 2013, last accessed 2nd October 2013

Sunstein C, Thaler R, (2003), *Libertarian Paternalism Is Not an Oxymoron*, The University of Chicago Law Review, Vol 70, Fall 2003, No 4, 1159, 1202

Thaler, R, H, (2002), *Save More Tommorow – A Simple Plan to Increase Retirement Saving*, Capital Ideas, Research Highlights from the University of

Chicago Graduate School of Business, Vol 4, No 1, Summer 2002

Thaler, R, H, Benartzi, S, (2001), *Save More Tomorrow: Using Behavioral Economics to Increase Employee Saving*, Working paper, The Anderson School at UCLA, August 2001

Thaler R, H, Sunstein C, (2003), *Libertarian Paternalism*, American Economic Review, 93, 2, 175-179

Thaler R, H, Sunstein, C, (2008), *Nudge – Improving Decisions About Health, Wealth and Happiness*, Yale University Press, New Haven

Thaler R, H, Sunstein, C, (2008), *Disclosure Is the Best Kind of Credit Regulation*, The Wall Street Journal, August 13th, 2008, 12:00 pm GMT, last accessed 17th February, 2009, printed in the paper version of the newspaper, page A17

Thaler R, H, Sunstein, C, (2009), *Preferences, Paternalism and Liberty*, Philosophy, Supplement, 59, 233-264

Toderov, A, Mandisodza, A, N, Goren, A, Hall, C, C, (2005), *Inferences of Competence from Faces Predict Election Outcomes*, Science, 308, 1623-26

Verbeek, B, Morris, C, (2010), *Game Theory and Ethics*, in *Standford Encyclopedia of Philosophy*, http://plato.stanford.edu/entris/game-ethics, last accessed 23 September, 2011.

Vicente (2006) quoted in *Nudge*

Voloch, E, (2003), *The Mechanism of the Slippery Slope*, Harvard Law Review, Vol 116, 1028-1136

Von Neumann, J, Morgenstern, O, (1944), *The Theory of Games and Economic Behavior*, Princeton University Press, Princeton, NY, USA

Vreind, N, J, (1995), The Self-Organization of Markets: An Example of a Computational Approach, Computational Economics, 8, 3, 205-31

Wall, S, (2008), *Perfectionism in Moral and Political Philosophy*, The Stanford Encyclopedia of Philosphy, Fall 2008 Edition, Zalta, E, editor,

URL=http://plato.stanford.edu/archives/fall2008/entries/perfectionism-moral

last accessed 28 November 2011

Wexler, B E, (2006), *Brain and Culture: Neurobiology, Ideology and Social Change*, MIT Press, Cambridge, MA

Whitman, G, (2006), *Against the New Paternalism: Internalities and the Economics of Self-Control*, Policy Analysis, Cato Institute, February, 22, 1-16

Whitman, G, (2010), *The Rise of New Paternalism*, Lead Essay, April 5th, 2010, www.cato-unbound.org/2010/04/05/glen-whitman/the-rise-of-the-new-paternalism, last accessed 2nd June 2010

Whitmore, J, (2011), personal communication

Zak, P, (2008), Editor, *Moral Markets - The Critical Role of Values in the Economy*, Princeton University Press, Princeton, New Jersey